Phoenix: Verifiable Post-Quantum Voting with Malicious Ballot-Box Privacy

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Abstract. E-voting systems must balance the voters' need for privacy with the election's goal of fairness and transparency, especially in a challenging and growing context like the post-quantum settings. While there are numerous protocol and systems with good voting guarantees under classical security setting (i.e., not post-quantum), there is a lack of substantial results for post-quantum schemes. All post-quantum e-voting protocols do not achieve satisfactory privacy - mostly due to informal or weak guarantees, while some offer a form of verifiability. Moreover, most schemes – even those specifically designed for online voting, do not offer the possibility of revoting.
In this work, we propose Phoenix, the first post-quantum voting system that is both verifiable and strongly private (i.e., private even when the adversary controls the ballot box). As we design our scheme based on Fully Homomorphic Signatures, we further address a gap in the design of this cryptographic primitive by proposing an efficient and context-hiding instantiation; previous constructions are either efficient or context-hiding (due to the heavy use of trapdoors). We enhance our contribution by providing an implementation of Phoenix via standard BFV encryption in Microsoft SEAL library and IBM Research's LaZer library for proof of knowledge and benchmark Phoenix on different CPUs.

1 Introduction

The idea of electronic voting is to modernize democratic processes through more accessibility and efficiency [1]; however, apart from the fact that people trust more in simple election systems with easily understandable procedures than complicated ones [2], designing voting schemes that can meet a wide range of security and privacy requirements remains one of the challenging problems in cryptography [3]. The paradox lies between transparency toward public verifiability and a high level of privacy for individual votes [2], as the level of privacy decreases by increasing verifiability and vice versa. This challenge gets more complicated by the advent of quantum computers, which will be in a position to retroactively break today's classically encrypted ballots [4].

Transition to post-quantum (PQ) e-voting needs to happen as soon as possible, because of the Store Now, Decrypt Later attack [5], but migration from classical cryptosystems to post-quantum ones is not trivial and all applications face many issues for this transition [6], from the substantial increase in key and ciphertext sizes imposed by post-quantum schemes [7] to interoperability and backward compatibility challenges arising during the coexistence of classical and quantum-resistant algorithms [8]. However, in voting systems, apart from these issues, this migration has even extra challenges, as mentioned earlier on balancing privacy and verifiability.

It is expected a voting system to have privacy [9] (no entity can learn the content of ballots) and verifiability [10–13] (voters can verify their ballots were recorded and anyone can verify correct tallying), so the current post-quantum electronic voting schemes need

to go beyond innovations in building block and consider bigger picture by taking account formal cryptographic privacy and verifiability properties in voting, because elections are more than just counting ballots [1]. No matter how much novel and efficient a new building block for voting scheme is, it will not be securely-constructive if the resulting voting system can not satisfy cryptographic privacy definitions (e.g., ballot privacy [9]). While studying post-quantum voting schemes [4,14–23], we observed that if a traditional voting system is ballot-private, it is not straightforward to instantiate a post-quantum variant of it [4], and just replacing its classical building blocks with post-quantum ones does not necessarily make the resulting voting system ballot-private. This is because, even when retaining the same architecture, migrating to post-quantum primitives (I) alters the underlying computational assumptions, thereby affecting information flow, privacy guarantees, and trust models [14]; and (II) introduces building blocks that, despite preserving classical properties (e.g., homomorphism), may possess additional attributes [4] capable of leaking information, e.g., the design ideas of discrete log-based voting protocols do not carry over easily to the lattice setting, due to specific problems [24] such as noise growth.

As summarized in Table 1, our study reveals a consistent pattern: no existing scheme simultaneously achieves strong ballot privacy, formal verifiability, and practical efficiency under post-quantum assumptions. Revoting support [25], important for usability and coercion mitigation, is similarly absent from all but one [19] scheme.

As part of our endeavor to fill aforementioned gaps by designing a new voting system, while we were using Fully Homomorphic Signature (FHS) in it, we noticed that the existing FHS miss a balance of efficiency and context-hiding property (revealing no info about signed data while remaining verifiable); Gorbunov et al.'s context-hiding FHS [26] requires $O(N)$ trapdoor samples per N-element dataset, whereas Luo et al.'s efficient construction [27] eliminates this cost but lacks context-hiding. No existing FHS provides both, yet without context-hiding a derived tally signature may leak individual ballot information.

1.1 Contributions

To address the gaps identified above, in this work we build Phoenix, the first PQ verifiable voting scheme that achieves delay-use malicious-ballot-box privacy [28], together with strong correctness[1], strong consistency, ideal functionality [9], and strong verifiability (individual, universal, and eligibility) [12], all under standard lattice assumptions, while also supporting revoting. As part of the design of Phoenix, we further construct the first efficient leveled fully homomorphic signature scheme with context-hiding, by building on the efficient evaluation of Luo et al. [27] and incorporating a context-hiding mechanism via the SampleLeft–SampleRight paradigm of Gorbunov et al. [26]. Eventually, we provide an implementation of Phoenix [29] and benchmark ballot creation across four platforms to demonstrate that end-to-end ballot creation completes in under 70 ms even for elections with up to 2^{25} candidates.

1.2 Related Works

The voting systems that are resistant to quantum attacks have diverse structures with minimal privacy and/or verifiability guarantees, where Table 1 summarizes our study.

[1] Individual verifiability: Voter confirms ballot was recorded. Universal verifiability: Anyone can verify correct tallying [10–13]. Strong correctness: Honest ballots are never rejected [9]. Strong consistency: Valid ballots tally correctly.

As a note, there are two main methods of election tally [30,31]: *Homomorphic* aggregates encrypted votes directly into an encrypted sum that is decrypted once, while *mix-net* anonymizes individual ciphertexts through a shuffle before decrypting each ballot.

Starting with Epoque [14], it splits each vote into secret shares, commits to each share, encrypts the shares under an IBE scheme (binding them to the voter's identity) and sends them to multiple trustees, who tally via homomorphic addition of commitments verified by a ZKP of well-formedness; all trustees must be online and actively participate in both the ballot weeding and tallying phases. Epoque's privacy proof uses a privacy definition called *Vote Privacy* [32] that Bernhard et al. [9] shows this is a weak definition of ballot privacy since it restricts the class of protocols, may yield inconsistent results under natural result functions (e.g., majority [33]), fails to apply to protocols outputting auxiliary data (e.g., tally proofs in ElectionGuard [34]), and misses privacy violations beyond same-vote detection [9]. Moreover, their Lemma 5 [31] omits the full hybrid argument for secret-share uniformity under commitment hiding. Also, Cortier et al. in [12] shows why the Epoque's verifiability definition [13] is weak, since it does not distinguish ballot deletion from ballot modification [12]. Moreover, concrete post-quantum parameters for all building blocks are also absent in Epoque.

Hough et al. proposed NTRU Mix-Net [15], in which votes are NTRU-encrypted, anonymized through a verifiable mix-net, and recovered via threshold distributed decryption, with ZKPs ensuring honest execution at each stage. In this scheme, NTRU parameters must be carefully chosen to avoid the overstretched regime where $q > 0.0058 \cdot \sigma^2 \cdot d^{2.484}$ (with d the polynomial degree, q the modulus, and σ the noise distribution width) enables subfield lattice attacks, and the scheme's privacy fundamentally relies on the assumption that at least one mix server remains honest without providing detection mechanisms for malicious mixing beyond proof verification. It is not straightforward how the privacy claims in [15] are achievable; although Hough et al. [15] does not propose a complete voting scheme, upon using their NTRU Mix-Net, they refer the privacy of resulting system to another mix-net by Aranha et al. [24], but even their theoretical system (full version's Appendix H [30]) does not provide any proofs or formal guarantees.

In FOO-based Voting [4], a voter encrypts their vote, gets a threshold blind signature on it from multiple authorities, and anonymously posts it to a bulletin board, where authorities later reveal decryption shares so anyone can tally. It combines threshold lattice-based blind signatures [35–37] with Dual-Regev encryption [38,39], but one-more unforgeability of the underlying blind signature remains conjectural [40], and privacy breaks if the required perfectly anonymous submission channel is compromised.

Code-based Voting [16], the first code-based verifiable voting system, relies on the less-studied dual-LPN assumption whose cryptanalytic background is limited, concrete 128-bit parameters against ISD attacks remain uncertain [41], and ballot privacy claims lack formal proofs.

In EVOLVE [17], each vote is split into additive shares, committed to multiple authorities using a lattice-based homomorphic commitment, with an OR-proof ensuring vote validity and amortized ZKPs by authorities enabling homomorphic tallying; it is post-quantum secure but lacks individual verifiability, requires a large voter threshold for efficiency, and supports no revoting. EVOLVE modifies the BPRIV definition [9] but the adaptation sidesteps the simulator requirement by making the tally consistent with the board in both worlds; in standard BPRIV, $\beta = 1$ requires a simulator Sim to produce a fake proof π' consistent with BB_1, which EVOLVE never constructs, and whether their notion implies standard BPRIV is not formally analyzed. EVOLVE also proves only a

Scheme	Assumptions	Building Blocks	Privacy	Verifiability	Revoting[2]
Epoque [14]	RLWE, RSIS	IBE [43], Com [44], NIZK [45]	Weak BPRIV [32]	indiv, univ	×
NTRU-Mix [15]	RLWE, NTRU	PKE [46], Com [47], ZKP [48–50]	×	univ	×
FOO-based [4]	RLWE, RSIS	BSign [35–37], TPKE [38,39]	×	partial univ	×
Code-based [16]	Dual-LPN	Com [16], NIZK [16,51], PKE [52,53]	×	indiv, univ	×
EVOLVE [17]	MLWE, MSIS	ZKP [54], Sig [55], Com [56], PKE [17]	BPRIV [9]	partial univ	×
EVOLVED [18]	MLWE, MSIS	Com [17], HW-Proof [57], PKE [18]	×	partial univ	×
Homo-LWE [19]	LWE	FHE [58], Sign [59], NIZK [60]	BPRIV [9]	indiv, univ	✓
BPRIV-BC [20]	MLWE, MSIS	Com [61], PostSec [62], NIBSign [63]	BPRIV [9]	indiv, univ	×
Kyber-BC [21]	MLWE	KEM [64], Block [65,66], Hash [67]	×	partial indiv	×
Hybrid-BC [22]	ECDH	ECDH [68], ABE [22], Block [69,70]	×	partial indiv	×
Steg-MQPC [23]	MQ	MQPC [71,72], LSB-Steg [23]	×	×	×
Phoenix **[Ours]**	RLWE, SIS	FHE **[73]**, ZKP **[74]**, FHS **[26,27]**	du-mb-BPRIV [28]	**indiv, univ**	✓

Table 1: Comparison of post-quantum electronic voting schemes based on cryptographic privacy properties with formal proofs providing and features they guarantee. du-mb-BPRIV: the ballot box cannot learn individual votes even when it is malicious and verification happens after tallying. Here, consider zero-knowledge proof ZKP and non-interactive one as NIZK, hamming weight proof HW-Proof, multivariate quadratic polynomial cryptography MQPC, identity-based encryption IBE, commitment Com, public-key encryption PKE, blind signature BSign, threshold public-key encryption TPKE, digital signature Sig, fully homomorphic encryption FHE, polynomial commitment PCom, posterior security PostSec, non-interactive blind signature NIBSign, key encapsulation mechanism KEM, blockchain Block, hash function Hash, elliptic-curve Diffie–Hellman ECDH, attribute-based encryption ABE, least-significant-bit steganography LSB-Steg, and fully homomorphic signature FHS.

weaker "realistic result" consistency rather than strong consistency [9]. EVOLVED [18] extends EVOLVE with Hamming Weight Proofs but inherits the same definitional gaps.

Chillotti et al. [19] present a homomorphic LWE-based voting scheme but provide only a proof sketch for ballot privacy. Abapour et al. [20] give a complete ballot privacy proof, but the scheme is exclusively for economic structures and, due to its blockchain dependency, lacks the flexibility for general use [42]; the same limitation applies to the blockchain-based schemes of Sekar et al. [21] and Mahanayak et al. [22] as well.

2 Preliminaries

Throughout this work, we use standard notation as λ denotes the security parameter; $[n]$ denotes the set $\{1,...,n\}$; $\|\cdot\|$ and $\|\cdot\|_\infty$ denote the Euclidean and infinity norms of a vector, respectively; $\widetilde{\mathbf{T}}$ denotes the Gram–Schmidt orthogonalization of a matrix $\mathbf{T}$.

2.1 Computational Assumptions

Let n,m,q be positive integers with q prime, $R_q = \mathbb{Z}_q[X]/(f(X))$ for an irreducible f of degree n, and χ a distribution over $\mathbb{Z}_q$.

SIS and MSIS [75, 76]. Given $\mathbf{A} \leftarrow\$ \mathbb{Z}_q^{n\times m}$, $\mathsf{SIS}_{n,m,q,\beta}$ asks to find a non-zero $\mathbf{z} \in \mathbb{Z}^m$ with $\mathbf{Az} = \mathbf{0} \pmod q$ and $\|\mathbf{z}\| \le \beta$. Its module variant $\mathsf{MSIS}_{n,k,m,q,\beta}$ asks for a non-zero $\mathbf{z} \in R_q^m$ with $\mathbf{Az} = \mathbf{0}$ in R_q^k and $\|\mathbf{z}\| \le \beta$, given $\mathbf{A} \leftarrow\$ R_q^{k\times m}$.

RLWE and MLWE [77, 78]. $\mathsf{RLWE}_{n,q,\chi}$ asks to distinguish $\{(a_i, a_i s + e_i)\}$ from $\{(a_i, u_i)\}$ for $a_i, u_i \leftarrow\$ R_q$, $s \leftarrow\$ R_q$, $e_i \leftarrow\$ \chi$. Its module variant $\mathsf{MLWE}_{n,k,\ell,q,\chi}$ asks to distinguish $(\mathbf{A}, \mathbf{As}+\mathbf{e})$ from $(\mathbf{A}, \mathbf{u})$ for $\mathbf{A} \leftarrow\$ R_q^{k\times\ell}$, $\mathbf{s} \leftarrow\$ \chi^\ell$, $\mathbf{e} \leftarrow\$ \chi^k$, $\mathbf{u} \leftarrow\$ R_q^k$.

[2] A *revote policy* Policy is a function that, given a list of ballots, selects which ballot to count per voter. Two common instantiations are: (I) only the first valid ballot cast by each voter is retained, (II) only the last valid ballot cast by each voter is retained (which Phoenix focuses on it).

2.2 Building Blocks

We now state the cryptographic primitives that our Phoenix builds upon.

Definition 1 (Hash Function [79]). *A cryptographic hash function* $H:\{0,1\}^* \to \{0,1\}^\lambda$ *is a deterministic function that maps arbitrary-length inputs to fixed-length outputs of* λ *bits, satisfying:*

- *For any PPT adversary* $\mathcal{A}$, $\Pr[(x,x') \leftarrow \mathcal{A}(1^\lambda) : x \neq x' \wedge H(x) = H(x')] \leq negl(\lambda)$ *(Collision Resistance).*
- *For random* $y \in \{0,1\}^\lambda$ *and PPT adversary* $\mathcal{A}$, $\Pr[x \leftarrow \mathcal{A}(y) : H(x) = y] \leq negl(\lambda)$ *(Preimage Resistance).*

Definition 2 (BFV Encryption [73]). *The* BFV *scheme is a leveled homomorphic encryption scheme whose security relies on the hardness of the* RLWE *problem.* BFV = (KeyGen,Enc,Dec,Add,Rerand) *consists of:*

- $(pk,sk) \leftarrow \mathsf{KeyGen}(1^\lambda)$*: Outputs public encryption key* pk *and secret decryption* sk*.*
- $c \leftarrow \mathsf{Enc}(pk,\mu)$*: Encrypts plaintext* μ *under* pk *and outputs ciphertext* c*.*
- $\mu \leftarrow \mathsf{Dec}(sk,c)$*: Decrypts ciphertext* c *using* sk *and recovers plaintext* μ*.*
- $c \leftarrow \mathsf{Add}(c_1,c_2)$*: Homomorphically adds two ciphertexts, which satisfies the relation of* $\mathsf{Dec}(sk,\mathsf{Add}(c_1,c_2)) = \mathsf{Dec}(sk,c_1) + \mathsf{Dec}(sk,c_2)$*.*
- $c' \leftarrow \mathsf{Rerand}(pk,c)$*: Re-randomises* c *to produce a fresh ciphertext* c' *satisfying* $\mathsf{Dec}(sk,c') = \mathsf{Dec}(sk,c)$*, indistinguishable from a fresh encryption of same plaintext.*

Based on [80, 81], BFV satisfies: *correctness* ($\mathsf{Dec}(sk,\mathsf{Enc}(pk,\mu)) = \mu$ with overwhelming probability for all $\mu \in R_t$, provided noise stays below the decryption threshold); *IND-CPA security* under the $\mathsf{RLWE}_{n,q,\chi_{\mathsf{err}}}$ ($|\Pr[\mathcal{A}(pk,\mathsf{Enc}(pk,\mu_0))=1] - \Pr[\mathcal{A}(pk,\mathsf{Enc}(pk,\mu_1))=1]| \leq \mathsf{negl}(\lambda)$ for all PPT $\mathcal{A}$, $\mu_0,\mu_1 \in R_t$); and BFV satisfies the *additive homomorphism* ($\mathsf{Dec}(sk,\mathsf{Add}(\mathsf{Enc}(pk,\mu_1),\mathsf{Enc}(pk,\mu_2))) = \mu_1 + \mu_2 \pmod t$).

Definition 3 (Lattice-Based Zero-Knowledge Proof System [74]). *The proof system* ZKP*, based on the hardness of* MSIS *and* MLWE*, proves knowledge of a witness satisfying a specified relation* $\mathcal{R}$*.* ZKP = (Setup,Prove,Verify) *consists of:*

- $pp \leftarrow \mathsf{Setup}(1^\lambda)$*: Outputs public parameters* pp *defining the relation* $\mathcal{R}$*.*
- $\pi \leftarrow \mathsf{Prove}(pp,x,w)$*: Given a statement* x *and witness* w *such that* $(x,w) \in \mathcal{R}$*, outputs a proof* π *via the Fiat–Shamir transform.*
- $\{0,1\} \leftarrow \mathsf{Verify}(pp,x,\pi)$*: Accepts if* π *is a valid proof that the prover knows a witness* w *such that* $(x,w) \in \mathcal{R}$*, and rejects otherwise.*

Here, proved relation is $\mathcal{R}_{\mathsf{vote}} = \{((c,pk,upk),(v,\mathbf{s}_1,\mathbf{e},\mathbf{s}_2)) \mid c = \mathsf{BFV.Enc}(pk,v;\mathbf{s}_1,\mathbf{e},\mathbf{s}_2) \wedge v \in \{0,1\} \wedge \|\mathbf{s}_1\|,\|\mathbf{e}\|,\|\mathbf{s}_2\| \leq \beta_{\mathsf{enc}}\}$, where the witness is the secret bit v and the randomness-error triple $(\mathbf{s}_1,\mathbf{e},\mathbf{s}_2)$ bounded by the BFV noise parameter β_{enc}.

Via the Fiat–Shamir transform, all ZKPs are non-interactive (secure in ROM). Considering [74, 82–84], proof system satisfies: *completeness* (Verify accepts an honest proof for any valid witness $(\mathbf{s}_1,\mathbf{m},\mathbf{s}_2)$ w.o.p.); *knowledge soundness* under MSIS (any PPT adversary producing an accepting proof for a false statement yields, via the ABDLOP extractor [74], a short vector breaking MSIS with non-negligible probability); and *computational zero-knowledge* under MLWE (a PPT simulator produces transcripts computationally indistinguishable from real proofs, so π leaks nothing beyond the truth of the asserted relations).

Definition 4 (Fully Homomorphic Signature). *Let $\ell, d_{\max} \in \mathbb{N}$ denote the maximum dataset size, and maximum supported circuit depth, respectively, and let $\mathcal{C}$ be a class of boolean circuits of depth at most $d_{\max}$ over* NAND *(universal boolean gates). The* leveled fully homomorphic signature FHS *consists of PPT algorithms as:*

- $(pk, sk) \leftarrow \mathsf{Setup}(1^\lambda, 1^{|\tau|}, 1^\ell, 1^{d_{\max}})$*: Given security parameter and bounds on tag length, dataset size, circuit depth, outputs a public key pk and a signing trapdoor sk.*
- $vk \leftarrow \mathsf{KeyGen}(pk, \tau)$*: Given pk and a dataset tag $\tau \in \{0,1\}^{|\tau|}$, derives a per-dataset verification key vk.*
- $\sigma_i \leftarrow \mathsf{Sign}(sk, vk, \tau, \mu_i, i)$*: Given trapdoor sk, verification key vk, tag τ, message bit $\mu_i \in \{0,1\}$, index $i \in [\ell]$, outputs signature σ_i authenticating dataset's i-th entry.*
- $\sigma_f \leftarrow \mathsf{Eval}(vk, \tau, \boldsymbol{\mu}, \boldsymbol{\sigma}, f)$*: Given vk, tag τ, message vector $\boldsymbol{\mu} \in \{0,1\}^\ell$, base signatures $\boldsymbol{\sigma}$, and a circuit $f \in \mathcal{C}$, outputs an evaluated signature σ_f authenticating $f(\boldsymbol{\mu})$.*
- $\{0,1\} \leftarrow \mathsf{Verify}(vk, \tau, \sigma_f, f)$*: Accepts if σ_f is a valid signature on $f(\boldsymbol{\mu})$ under tag τ, and rejects otherwise.*
- $\tilde{\sigma}_f \leftarrow \mathsf{Hide}(vk, y, \sigma_f)$*: Given vk, output value $y = f(\boldsymbol{\mu})$, evaluated signature σ_f, re-randomizes σ_f into context-hiding signature $\tilde{\sigma}_f$ that reveals nothing beyond y.*
- $\{0,1\} \leftarrow \mathsf{HVerify}(vk, \tau, \tilde{\sigma}_f, y, f)$*: Accepts if $\tilde{\sigma}_f$ is a valid context-hiding signature certifying that f evaluated to y under tag τ, and rejects otherwise.*

Based on [26,27,85], we consider the following properties for FHS with a focus on unforgeability and context-hiding.

Definition 5 (Signing Correctness). *We say that the* FHS *scheme is* signing-correct *if for all tags $\tau = \{\tau_0, \ldots, \tau_{|\tau|-1}\} \in \{0,1\}^{|\tau| \geq \lambda}$, all messages $\mu_i \in \{0,1\}$, and all indices $i \in [\ell]$,*

$$\Pr\left[\begin{array}{l} (pk, sk) \leftarrow \mathsf{Setup}(1^\lambda, 1^{|\tau|}, 1^\ell, 1^{d_{\max}}); \\ vk \leftarrow \mathsf{KeyGen}(pk, \tau); \\ \sigma_i \leftarrow \mathsf{Sign}(sk, vk, \tau, \mu_i, i) \end{array} : \mathsf{Verify}(vk, \tau, \sigma_i, \mathsf{id}_i) = 1\right] = 1,$$

where id_i denotes the projection circuit $\mathsf{id}_i(\mu_1, \ldots, \mu_\ell) = \mu_i$ and the probability is over the coins of the algorithms.

Definition 6 (Evaluation Correctness). *We say that the* FHS *scheme is* evaluation-correct *if for all tags $\tau \in \{0,1\}^{|\tau| \geq \lambda}$, all message vectors $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_\ell) \in \{0,1\}^\ell$, and all admissible circuits $f \in \mathcal{C}$ of depth at most $d_{\max}$,*

$$\Pr\left[\begin{array}{l} (pk, sk) \leftarrow \mathsf{Setup}(1^\lambda, 1^{|\tau|}, 1^\ell, 1^{d_{\max}}); \\ vk \leftarrow \mathsf{KeyGen}(pk, \tau); \\ \sigma_i \leftarrow \mathsf{Sign}(sk, vk, \tau, \mu_i, i) \text{ for } i \in [\ell]; \\ \sigma_f \leftarrow \mathsf{Eval}(vk, \tau, \boldsymbol{\mu}, \boldsymbol{\sigma}, f) \end{array} : \mathsf{Verify}(vk, \tau, \sigma_f, f) = 1\right] = 1,$$

where the probability is over the coins of the algorithms. Moreover, correctness extends to composed evaluation: for circuits $h_1, \ldots, h_s$ with $h_j : \{0,1\}^\ell \to \{0,1\}$ and $g : \{0,1\}^s \to \{0,1\}$, if (vk, σ_j) satisfies $\mathsf{Verify}(vk, \tau, \sigma_j, h_j) = 1$ for each $j \in [s]$, then $\sigma^* \leftarrow \mathsf{Eval}(vk, \tau, (h_1(\boldsymbol{\mu}), \ldots, h_s(\boldsymbol{\mu})), (\sigma_1, \ldots, \sigma_s), g)$ satisfies $\mathsf{Verify}(vk, \tau, \sigma^*, g \circ \bar{h}) = 1$, where $(g \circ \bar{h})(\boldsymbol{\mu}) = g(h_1(\boldsymbol{\mu}), \ldots, h_s(\boldsymbol{\mu}))$.

Definition 7 (Succinctness). *We say that the* FHS *scheme is* succinct *if for the security parameter λ, any tag $\tau \in \{0,1\}^{|\tau| \geq \lambda}$, all message vectors $\boldsymbol{\mu} \in \{0,1\}^\ell$, and all*

admissible circuits $f \in \mathcal{C}$,

$$\Pr\left[\begin{array}{l}(pk,sk) \leftarrow \mathsf{Setup}(1^\lambda,1^{|\tau|},1^\ell,1^{d_{\max}});\\ vk \leftarrow \mathsf{KeyGen}(pk,\tau);\\ \sigma_i \leftarrow \mathsf{Sign}(sk,vk,\tau,\mu_i,i) \text{ for } i \in [\ell];\\ \sigma_f \leftarrow \mathsf{Eval}(vk,\tau,\boldsymbol{\mu},\boldsymbol{\sigma},f)\end{array} : \|\sigma_f\| \le p(\lambda)\cdot\log\ell\right] = 1,$$

where $p(\lambda)$ *is a fixed polynomial in the security parameter. In particular, the size of the evaluated signature* σ_f *depends at most logarithmically on the dataset size* ℓ *and is independent of the circuit size of* f.

Definition 8 (Existential Unforgeability under Fully Adaptive Chosen-Message Attacks). *We define the security of the* FHS *scheme via the following experiment* $\mathsf{Expt}_{\mathcal{A}}^{\text{EU-F-ACMA}}(1^\lambda)$ *between a challenger and a PPT adversary* $\mathcal{A}$*:*

1. ***Setup.*** *The challenger runs* $(pk,sk) \leftarrow \mathsf{Setup}(1^\lambda,1^{|\tau|},1^\ell,1^{d_{\max}})$ *and sends* pk *to* $\mathcal{A}$.
2. ***Signing Queries.*** *The adversary* $\mathcal{A}$ *adaptively submits queries of the form* (τ,μ_i,i), *where the tag* $\tau \in \{0,1\}^{|\tau|\ge\lambda}$ *is a dataset identifier,* $\mu_i \in \{0,1\}$, *and* $i \in [\ell]$. *The challenger maintains a list* T_τ *for each queried tag* τ *and responds with* $\sigma_i \leftarrow \mathsf{Sign}(sk,vk,\tau,\mu_i,i)$, *adding* (μ_i,i) *to* T_τ *(aborting if* $|T_\tau| > \ell$ *or if a different message was previously signed at the same index).*
3. ***Forgery.*** *The adversary* $\mathcal{A}$ *outputs a tuple* $(\tau^*,\sigma^*_{f^*},\mu^*,f^*)$. *The experiment outputs* 1 *if and only if* $\mathsf{Verify}(vk^*,\tau^*,\sigma^*_{f^*},f^*)=1$ *(or* $\mathsf{HVerify}(vk^*,\tau^*,\tilde{\sigma}^*_{f^*},\mu^*,f^*)=1$ *when context hiding is applied) and one of the following holds:*
   - Type I forgery: $\tau^* \neq \tau_j$ *for any previously queried tag* τ_j.
   - Type II forgery: $\tau^* = \tau_j$ *for some queried tag* τ_j, *and* $\mu^* \neq f^*(\boldsymbol{\mu}_j)$, *where* $\boldsymbol{\mu}_j$ *is the message vector signed under* τ_j.

We say that the FHS *scheme is* existentially unforgeable against fully adaptive chosen-message attacks *if for all PPT adversaries* $\mathcal{A}$, $\Pr[\mathsf{Expt}_{\mathcal{A}}^{\text{EU-F-ACMA}}(1^\lambda)=1] \le \mathsf{negl}(\lambda)$.

Definition 9 (Context-Hiding Security). *We say that the* FHS *scheme supports* statistical context hiding *if there exist additional PPT procedures* $\tilde{\sigma}_f \leftarrow \mathsf{Hide}(vk,y,\sigma_f)$ *and* $\mathsf{HVerify}(vk,\tau,\tilde{\sigma}_f,y,f)$ *such that the following three properties hold:*

1. ***Correctness of Hiding.*** *For any* $(pk,sk) \in \mathsf{Setup}(1^\lambda,1^{|\tau|},1^\ell,1^{d_{\max}})$, $vk \in \mathsf{KeyGen}(pk,\tau)$, *and any* (σ_f,y,f) *such that* $\mathsf{Verify}(vk,\tau,\sigma_f,f)=1$, *for any* $\tilde{\sigma}_f \in \mathsf{Hide}(vk,y,\sigma_f)$ *we have* $\mathsf{HVerify}(vk,\tau,\tilde{\sigma}_f,y,f)=1$.
2. ***Unforgeability.*** *The unforgeability guarantee of Definition 8 continues to hold when the* Verify *procedure in the security experiment is replaced by* HVerify.
3. ***Statistical Indistinguishability.*** *There exists a PPT simulator* Sim *such that for any fixed (worst-case) choice of* $(pk,sk) \in \mathsf{Setup}(1^\lambda,1^{|\tau|},1^\ell,1^{d_{\max}})$, $vk \in \mathsf{KeyGen}(pk,\tau)$, *and any* (σ_f,y,f) *satisfying* $\mathsf{Verify}(vk,\tau,\sigma_f,f)=1$, *we have* $\mathsf{Hide}(vk,y,\sigma_f) \overset{\text{stat}}{\approx} \mathsf{Sim}(sk,vk,\tau,y,f)$, *where the randomness is only over the coins of* Hide *and* Sim. *In particular, the dataset tag component* τ *in the verification key* vk *depends only on* (sk,τ) *and not on the data* $\{\mu_i\}$.

3 Post-Quantum Verifiable **du-mb-BPRIV** Voting System

Entities. Our voting system, Phoenix, involves four distinct roles. A *Registrar* generates and privately distributes signing keys to voters, sending the corresponding verification keys to the server. *Voters* encrypt and sign their ballots, submitting them through an authenticated channel, with only their latest ballot retained if they re-vote. The *Voting Server* maintains the *Public Bulletin Board* (PBB), validating and recording all accepted ballots. Finally, a group of decryption *trustees* collectively hold the election's private key under a threshold scheme, so that only a required minimum number of them need to participate to decrypt the final result.

Voting Protocol. In Phoenix, voters encrypt their vote under the election public key, prove its well-formedness via a ZKP, and sign the resulting ballot with their credential key, after which the server validates and records it on the BB for homomorphic tallying. The Phoenix consists of the following procedures.

- Setup(1^λ): Runs $(pk,sk) \leftarrow \mathsf{BFV.KeyGen}(1^\lambda)$ and $pp \leftarrow \mathsf{ZKP.Setup}(1^\lambda)$. Returns the election key pair (pk,sk) and public parameters pp.
- Register(id): Runs $(pk_{fhs},sk_{fhs}) \leftarrow \mathsf{FHS.Setup}(1^\lambda,1^{|\tau|},1^\ell,1^{d_{\max}})$, draws a fresh per-voter tag $\tau \leftarrow\$ \{0,1\}^{|\tau|}$ independently of id, and sets $upk \leftarrow (\mathsf{FHS.KeyGen}(pk_{fhs},\tau),\tau)$ and $usk \leftarrow sk_{fhs}$. The tag is bundled into upk so that it passes de-identification and remains available on pbb for FHS.Verify[3].
- Vote(id,v,pk,upk,usk): The voter parses $upk=(vk,\tau)$, encrypts the vote as $c \leftarrow \mathsf{BFV.Enc}(pk,v)$, produces a proof $\sigma \leftarrow \mathsf{ZKP.Prove}(pp,(c,upk),(v,\mathbf{s}_1,\mathbf{e},\mathbf{s}_2))$ binding the statement to upk (preventing cross-voter replay), and signs the ciphertext–proof pair $s \leftarrow \mathsf{FHS.Sign}(usk,vk,\tau,(c,\sigma),1)$. The signature is over the ballot contents alone and is independent of id, so that ballots remain publicly verifiable on pbb after de-identification. Returns $b=(id,upk,c,\sigma,s)$; id appears only in the envelope, for the server's access-control check in Valid, and is removed by remID before publication.
- Valid(BB,b,pk): Parses $b=(id,upk,c,\sigma,s)$ and verifies three conditions: (1) no ballot in BB pairs id with a different credential, or uses upk for a different identity (i.e., each identity maps to exactly one credential, preventing double voting under different keys); (2) the zero-knowledge proof is valid: $\mathsf{ZKP.Verify}(pp,(c,upk),\sigma)=1$, ensuring the encrypted vote is well-formed; (3) signature is valid: $\mathsf{FHS.Verify}(vk,\tau,s,(c,\sigma))$, confirming the ballot was created by the legitimate credential holder. Returns true iff all three conditions hold.
- Publish(BB): Publishes each ballot together with its hash fingerprint: $pbb = \{(b,\ H(b)) \mid b \in C\}$. Voter identities are removed from published ballots to protect anonymity, where $\mathsf{remID}(b)$ removes the voter identity id from ballot $b=(id,upk,c,\sigma,s)$, yielding $\tilde{b}=(upk,c,\sigma,s)$, so $pbb=\{(\tilde{b},H(\tilde{b}))\,|\,\tilde{b}=\mathsf{remID}(b),b\in C\}$.
- Tally(BB,sk): The server first retains, for each credential upk, only its most recent entry in BB, yielding BB^*; this realises re-voting and bounds each voter's contribution to one ballot. It then verifies every $b \in BB^*$ with ZKP.Verify and

[3] Here, τ is the FHS dataset tag of Definition 4, instantiated per-voter; bundling it into upk makes sure it remains available on pbb (where id has been removed) for subsequent FHS.Verify calls in Valid and Tally.

FHS.Verify, collecting the ciphertexts of accepted ballots in vbb. Using the additive homomorphism of BFV, it computes $c^* \leftarrow \mathsf{BFV.Add}(vbb)$ and decrypts $r \leftarrow \mathsf{BFV.Dec}(sk,c^*)$. Finally, it derives the public view $pbb \leftarrow \mathsf{Publish}(BB)$ and produces a zero-knowledge proof of correct tallying $\pi \leftarrow \mathsf{ZKP.Prove}(pp, (pk, pbb, r), sk)$ certifying that r matches the published board. Returns (r,π).

- Verify$((pk,pbb,r),\pi)$: Checks the tally proof by running $\mathsf{ZKP.Verify}(pp,(pk,pbb,r),\pi)$ and returns its boolean output, allowing any third party to confirm that the published result r correctly reflects the contents of the public bulletin board pbb.
- VerifyVote(b,pbb): Allows a voter to confirm their ballot was correctly recorded by checking whether $H(b) \in \{h \mid (b',h) \in pbb\}$. Using a hash fingerprint rather than the full ballot makes the check lightweight, while collision resistance of H ensures the check is sound.

Note that sk and usk are treated as single secrets above; in practice, both can be threshold-shared among multiple trustees and registrars respectively [86], without affecting the security proofs. Phoenix supports re-voting as if a voter submits a new ballot, only the most recent one is retained on the bulletin board.

Phoenix uses FHS on singleton datasets ($\ell = 1$): a ballot is signed, then later verified, with no intermediate evaluation. For a message $M \in \{0,1\}^{\ell_M}$ we therefore write $s \leftarrow \mathsf{FHS.Sign}(usk, vk, \tau, M, 1)$ as shorthand for bit-wise signing of M followed by FHS.Eval on the identity circuit f_M that outputs M; and $\mathsf{FHS.Verify}(vk,\tau,s,M)$ for the matching call $\mathsf{FHS.Verify}(vk, \tau, s, f_M)$ of Definition 4; both calls unfold to the 5-input/4-input of Definition 4.

```
Setup(1^λ)
  (pk,sk) ← BFV.KeyGen(1^λ)
  pp ← ZKP.Setup(1^λ)
  return (pk,sk,pp)

Register(id)
  (pk_fhs,sk_fhs) ← FHS.Setup(1^λ,1^|τ|,1^ℓ,1^d_max)
  τ ←$ {0,1}^|τ|  // per-voter tag, independent of id
  upk ← (FHS.KeyGen(pk_fhs,τ),τ)
  usk ← sk_fhs
  return (upk,usk)

Verify((pk,pbb,r),π)
  return ZKP.Verify(pp,(pk,pbb,r),π)

VerifyVote(b,pbb)
  b̃ ← remID(b)
  return H(b̃) ∈ {h | (·,h) ∈ pbb}

Vote(id,v,pk,upk,usk)
  (vk,τ) ← upk
  c ← BFV.Enc(pk,v)
  σ ← ZKP.Prove(pp,(c,upk),(v,s_1,e,s_2))
  s ← FHS.Sign(usk,vk,τ,(c,σ),1)
  return b ← (id,upk,c,σ,s)

Valid(BB,b,pk)
  (id,upk,c,σ,s) ← b
  e_1 ← ∀x,y. (x,y,·,·,·) ∈ BB ⇒
    (x,y) = (id,upk) ∨ (x ≠ id ∧ y ≠ upk)
  e_2 ← ZKP.Verify(pp,(c,upk),σ)
  e_3 ← FHS.Verify(vk,τ,s,(c,σ))
  return (e_1 ∧ e_2 ∧ e_3)

Publish(BB)
  pbb ← {(b̃,H(b̃)) | b̃ ← remID(b)}
  return pbb

Tally(BB,sk)
  vbb ← []
  BB* ← {b ∈ BB | b is the latest entry for its upk}
  for b ∈ BB* do
    (id,upk,c,σ,s) ← b;  (vk,τ) ← upk
    e_1 ← ZKP.Verify(pp,(c,upk),σ)
    e_2 ← FHS.Verify(vk,τ,s,(c,σ))
    if (e_1 ∧ e_2) then vbb ← vbb ∪ {c}
  c* ← BFV.Add(vbb)
  r ← BFV.Dec(sk,c*)
  pbb ← Publish(BB)
  π ← ZKP.Prove(pp,(pk,pbb,r),sk)
  return (r,π)
```

Fig. 1: The Phoenix voting system built upon BFV = (KeyGen, Enc, Dec, Add), ZKP = (Setup,Prove,Verify), FHS = (Setup,KeyGen,Sign,Verify), and hash function H. The function remID(b) removes the voter identity from ballot b.

3.1 Concrete Parameters

Phoenix targets $\lambda = 128$ bits of post-quantum security throughout. H is instantiated with SHA-256 (256-bit output, ≥ 128-bit collision resistance, unaffected by Grover's

algorithm beyond a factor-two preimage reduction [87]). For BFV, we adopt the Homomorphic Encryption Standard [88] parameter set ($n=4096$, $\log q=109$, $t=2$), achieving $\approx$128-bit security under $\mathsf{RLWE}_{n,q,\chi}$ per the lattice estimator [89]. For ZKP, we follow the ABDLOP instantiation of [74] with $d=4$, $n=64$, $\log q\approx 32$, challenge space 2^{256}, and $\beta=2^{14}$, achieving 128-bit soundness under $\mathsf{MSIS}_{d,m,\beta'}$ and $\mathsf{MLWE}_{d,d,\chi}$. For FHS, we set $n=512$, $m=2n\lceil\log q\rceil\approx 22{,}528$, $\log q=22$, with verification bound B derived from $s\geq\|\widetilde{\mathbf{T}_{\mathbf{A}^*}}\|\cdot\omega(\sqrt{\log m})$ [90,91], context-hiding bound $\tilde{\beta}_{\max}=\beta_{\max}\cdot p_{\mathrm{extra}}(n,m,\log q)$ (Definition 10), and $d_{\max}=30$ sufficient for Phoenix's tallying circuits.

3.2 Performance and Efficiency Analysis

The core implementation of Phoenix is available in [29]; we benchmarked the Phoenix voting client, which encompasses BFV encryption, ZKP generation, and FHS signing, across four representative platforms: an Android Qualcomm Snapdragon 8 Gen 2 (aarch64) via PRoot Distro[4] on Termux, an Apple M2 (macOS), a Dell Precision 3581 equipped with an Intel Core i7-1355U (Windows 11), the same Intel Core i7-1355U with different RAM running Ubuntu 22.04. As shown in Table 2, ballot creation time is primarily determined by the vote-encoding length $k=\lceil\log_2 n_{cand}\rceil$ augmented by the fixed $\kappa=16$ credential bits, which shows that the per-ballot cost of Phoenix grows linearly in k. These sub-70 ms timings show Phoenix is practical even on mobile hardware for elections with up to 2^{25} candidates ($k=25$), with no inherent constraint on the number of voters as ballots are created independently at fixed per-voter cost.

	Dell Precision 3581	Apple Silicon M2	Intel Core i7-1355U	Snapdragon 8 Gen 2
2^1 Candidates	2.719×10^{-3}	2.368×10^{-3}	2.631×10^{-3}	2.894×10^{-3}
2^5 Candidates	11.817×10^{-3}	9.749×10^{-3}	10.931×10^{-3}	12.120×10^{-3}
2^{10} Candidates	22.099×10^{-3}	19.385×10^{-3}	20.737×10^{-3}	22.996×10^{-3}
2^{25} Candidates	58.713×10^{-3}	48.748×10^{-3}	54.841×10^{-3}	68.560×10^{-3}

Table 2: Time in terms of seconds (averaged over three runs) to encrypt, commit, and sign a ballot with a k-bit vote encoding (2^k candidates) and $\kappa=16$ credential bits.

4 Efficient Context-Hiding Leveled FHS

In Phoenix we use fully homomorphic signature but existing FHS constructions face a gap: Gorbunov et al. [26] sign each element independently via $\mathbf{U}_i\leftarrow\mathsf{SamPre}(\mathbf{A},\mathbf{V}_i-x_i\mathbf{G},\mathsf{td})$, requiring N costly trapdoor inversions, but achieve context-hiding by re-randomising the evaluated signature through SampleRight. Luo et al. [27] reduce this to just two SampleLeft calls by encoding both bit values via $\mathsf{encode}(k)=[k\mathbf{I}_n\,|\cdots|\,k\mathbf{I}_n]\mathbf{G}_n$ and deriving each σ_i deterministically through Unpack, but return the evaluated signature without re-randomisation, potentially leaking individual data beyond the computation output.

We resolve this by presenting the first FHS that achieves both properties simultaneously and instantiate the FHS in Phoenix with this variant. Since our scheme operates on multiple datasets distinguished by tags τ, we adopt the multi-data syntax: a signer can sign arbitrarily many datasets under different tags, each tag τ producing

[4] https://github.com/termux/proot-distro

a verification key vk via KeyGen, and the verifier must know the tag τ of the dataset on which the computation was performed.

4.1 Our Fully Homomorphic Signature Scheme

Built upon [27], our scheme uses *lattice trapdoors* $\mathsf{LT}=(\mathsf{GenTrap},\mathsf{SampleLeft},\mathsf{SampleRight})$ [90,91], where GenTrap outputs $(\mathbf{A}^*,\mathbf{T}_{\mathbf{A}^*})$, SampleLeft samples a short preimage under $[\mathbf{A}^*\,|\,\mathbf{C}]$, and SampleRight re-randomises a solution into a fresh statistically independent short preimage; *vector encoding* $\mathsf{VE}=(\mathsf{encode},\mathsf{Unpack})$, where $\mathsf{encode}(k)$ maps $k\in\{0,1\}$ to a structured offset matrix and $\mathsf{Unpack}(\mathbf{R},i)$ extracts the i-th block; and *tag evaluation* $\mathsf{TE}=(\mathsf{PubEval},\mathsf{TrapEval})$, where $\mathsf{PubEval}_{(\tau,\nu,t)}$ derives the per-tag matrix $\mathbf{B}_\tau\in\mathbb{Z}_q^{n\times m}$ from $(\mathbf{B},\mathbf{B}')$. The gadget matrix $\mathbf{G}_n\in\mathbb{Z}_q^{n\times m}$ [91] admits binary decomposition $\mathbf{G}_n^{-1}(\cdot)$, $\mathbf{P}_{x_i}$ is the permutation selecting the i-th gadget block, and the verification bound B is derived from $s\geq\|\widetilde{\mathbf{T}_{\mathbf{A}^*}}\|\cdot\omega(\sqrt{\log m})$.

Definition 10 (Efficient Context-Hiding Leveled Fully Homomorphic Signature). *Let $\lambda,\ell,d_{\max}\in\mathbb{N}$ be the security parameter, maximum dataset size, and maximum circuit depth. Let n,m,q be lattice parameters with $m\geq 2n\lceil\log q\rceil$ determined by $(\lambda,d_{\max})$ under $\mathsf{SIS}_{n,m,q,\beta}$, $|\tau|\geq\lambda$ the tag bit-length, and $\tilde{\beta}_{\max}=\beta_{\max}\cdot p_{\mathrm{extra}}(n,m,\log q)$ the context-hiding norm bound. The fully homomorphic signature scheme* $\mathsf{FHS}=(\mathsf{Setup},\mathsf{KeyGen},\mathsf{Sign},\mathsf{Eval},\mathsf{Verify},\mathsf{Hide},\mathsf{HVerify})$ *is given in Figure 2.*

$\mathsf{FHS.Setup}(1^\lambda,1^{|\tau|},1^\ell,1^{d_{\max}})$

$(\mathbf{A}^*,\mathbf{T}_{\mathbf{A}^*})\leftarrow\mathsf{GenTrap}(1^n,1^m,q)$
$\mathbf{A},\mathbf{D}_0,\mathbf{D}_1,\mathbf{B},\mathbf{B}'\leftarrow\$\,\mathbb{Z}_q^{n\times m}$
$\mathbf{r}_0\leftarrow\$\,\{0,1\}^m;\quad \mathbf{z}\leftarrow\mathbf{A}^*\cdot\mathbf{r}_0\in\mathbb{Z}_q^n$
$\nu\leftarrow\lceil\log\lceil\log|\tau|\rceil\rceil;\quad t\leftarrow\lceil 2\log 2|\tau|\rceil$
$pk\leftarrow(\mathbf{A},\mathbf{A}^*,\mathbf{D}_0,\mathbf{D}_1,\mathbf{B},\mathbf{B}',\mathbf{z},\nu,t)$
$sk\leftarrow\mathbf{T}_{\mathbf{A}^*}$
return (pk,sk)

$\mathsf{FHS.KeyGen}(pk,\tau)$

$\mathbf{B}_\tau\leftarrow\mathsf{PubEval}_{(\tau,\nu,t)}(\mathbf{B},\mathbf{B}',h_{\mathbf{z},\boldsymbol{\alpha},\boldsymbol{\beta}})$
$\mathbf{A}_\tau\leftarrow[\mathbf{A}^*\,|\,\mathbf{A}+\mathbf{B}_\tau]\in\mathbb{Z}_q^{n\times 2m}$
return $vk\leftarrow(\mathbf{A}_\tau,\mathbf{D}_0,\mathbf{D}_1,\mathbf{z})$

$\mathsf{FHS.Hide}(vk,y,\sigma_f)$

$(\mathbf{R}_f,y,f)\leftarrow\sigma_f$
Recompute $\mathbf{A}_\tau,\mathbf{D}_f$ from vk,f
$\mathbf{H}_f\leftarrow[\mathbf{A}_\tau\,|\,\mathbf{D}_f+(y-1)\mathbf{G}_n]\in\mathbb{Z}_q^{n\times 3m}$
return $\tilde{\sigma}_f\leftarrow\mathsf{SampleRight}(\mathbf{H}_f,\mathbf{R}_f,\mathbf{z})$

$\mathsf{FHS.Sign}(sk,vk,\tau,\mu_i,i)$

for $k\in\{0,1\}$ **do**
$\begin{bmatrix}\mathbf{R}_{k,1}\\ \mathbf{R}_{k,2}\end{bmatrix}\leftarrow\mathsf{SampleLeft}(\mathbf{A}^*,\mathbf{A}+\mathbf{B}_\tau,\mathbf{T}_{\mathbf{A}^*},\mathbf{D}_k+\mathsf{encode}(k),s)$
$\begin{bmatrix}\mathbf{R}_{\mu_i,1,i}\\ \mathbf{R}_{\mu_i,2,i}\end{bmatrix}\leftarrow\mathsf{Unpack}\left(\begin{bmatrix}\mathbf{R}_{\mu_i,1}\\ \mathbf{R}_{\mu_i,2}\end{bmatrix},i\right)$
return $\sigma_i\leftarrow\left(\begin{bmatrix}\mathbf{R}_{\mu_i,1,i}\\ \mathbf{R}_{\mu_i,2,i}\end{bmatrix},\mu_i,i\right)$

$\mathsf{FHS.Eval}(vk,\tau,\boldsymbol{\mu},\boldsymbol{\sigma},f)$

for each NAND-gate $g=(u,v,w)$ in f **do**
$\hat{\mathbf{D}}_u\leftarrow\mathbf{G}_n^{-1}(\mathbf{D}_u\mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n))$
$\mathbf{R}_w\leftarrow\mathbf{R}_v\hat{\mathbf{D}}_u-y\mathbf{R}_u$
$\mathbf{D}_w\leftarrow\mathbf{D}_v\mathbf{G}_n^{-1}(\mathbf{P}_{y_j}\mathbf{G}_n)\hat{\mathbf{D}}_u-\mathbf{G}_n$
return $\sigma_f\leftarrow(\mathbf{R}_f,f(\boldsymbol{\mu}),f)$

$\mathsf{FHS.Verify}(vk,\tau,\sigma_f,f)$

$(\mathbf{R}_f,y,f)\leftarrow\sigma_f$
Recompute $\mathbf{A}_\tau,\mathbf{D}_f$ from vk,τ,f
$e_1\leftarrow\|\mathbf{R}_f\|_\infty\leq B$
$e_2\leftarrow(\mathbf{A}_\tau\mathbf{R}_f=\mathbf{D}_f+y\mathbf{G}_n)$
return $(e_1\wedge e_2)$

$\mathsf{FHS.HVerify}(vk,\tau,\tilde{\sigma}_f,y,f)$

Recompute $\mathbf{A}_\tau,\mathbf{D}_f$ from vk,τ,f
$\mathbf{H}_f\leftarrow[\mathbf{A}_\tau\,|\,\mathbf{D}_f+(y-1)\mathbf{G}_n]$
$e_1\leftarrow(\mathbf{H}_f\cdot\tilde{\sigma}_f=\mathbf{z})$
$e_2\leftarrow(\|\tilde{\sigma}_f\|_\infty\leq\tilde{\beta}_{\max})$
return $(e_1\wedge e_2)$

Fig. 2: Our context-hiding leveled FHS, where a simulator $\tilde{\sigma}_f\leftarrow\mathsf{Sim}\ (sk,\ vk,\ \tau,\ y,\ f)$ produces output indistinguishable from Hide by the SampleLeft-SampleRight statistical closeness (see Lemma 6.4 of [26]). The matrix $\mathbf{G}_n$ denotes the gadget matrix and $B,\tilde{\beta}_{\max}$ the verification and context-hiding norm bounds, respectively. $\mathbf{R}_{k,1}$ and $\mathbf{R}_{k,2}$ are left and right preimage matrix blocks.

Due to space limitations, for other properties and complete formal proofs of Theorems in this work refer to Appendix of full version [92].

For complete formal proofs of Theorems refer to Appendix A.

Theorem 1. *The* FHS *scheme of Definition 10 satisfies signing correctness (Definition 5).*

Theorem 2. *The* FHS *scheme of Definition 10 satisfies evaluation correctness (Definition 6).*

Theorem 3. *The* FHS *scheme of Definition 10 satisfies succinctness (Definition 7).*

Theorem 4. *Under* $\mathsf{SIS}(n,m,q,\beta)$*, for every PPT adversary* $\mathcal{A}$*,* $\Pr\big[\mathsf{Expt}^{\text{EU-F-ACMA}}_{\mathcal{A},\mathsf{FHS}}(1^\lambda)=1\big] \leq \mathsf{Adv}^{\mathsf{sis}}_{n,m,q,\beta}(\lambda) + \mathsf{negl}(\lambda)$*, where* FHS *is the scheme of Definition 10 and* $\mathsf{Expt}^{\text{EU-F-ACMA}}$ *is the experiment of Definition 8.*

Any winning adversary must produce $(\tau^*,\sigma^*_{f^*},\mu^*,f^*)$ with $\mathsf{Verify}(vk^*,\tau^*,\sigma^*_{f^*},f^*)=1$. In a Type I forgery, the extracted $\mathbf{R}^*_f$ satisfies $\mathbf{A}_{\tau^*}\mathbf{R}^*_f=\mathbf{D}_{f^*}+\mu^*\mathbf{G}_n$ with $\|\mathbf{R}^*_f\|_\infty \leq B$ on a never-queried tag, yielding a short SIS solution directly. In a Type II forgery, subtracting the verification equations for two valid signatures on the same tag τ_j but distinct outputs produces a non-trivial short kernel vector, again breaking $\mathsf{SIS}_{n,m,q,\beta}$.

Theorem 5. *Let* $(pk,sk)\leftarrow\mathsf{Setup}(1^\lambda,1^{|\tau|},1^\ell,1^{d_{\max}})$*,* $vk\leftarrow\mathsf{KeyGen}(pk,\tau)$*, and suppose* $\sigma_f=(\mathbf{R}_f,y,f)$ *satisfies* $\mathsf{Verify}(vk,\tau,\sigma_f,f)=1$*. Then for any* $\tilde{\sigma}_f\leftarrow\mathsf{Hide}(vk,y,\sigma_f)$*, we have* $\mathsf{HVerify}(vk,\tau,\tilde{\sigma}_f,y,f)=1$*.*

Theorem 6. *Under the* $\mathsf{SIS}(n,m,q,\beta)$ *assumption with* $\beta_{\mathsf{SIS}}=2^{\omega(\log\lambda)}\cdot\beta^2_{\max}$*, the* FHS *scheme remains existentially unforgeable when the verification procedure in Definition 8 is replaced by* HVerify*.*

Theorem 7. *The* FHS *satisfies Definition 9: for every (unbounded) adversary* $\mathcal{A}$*, every* $(pk,sk)\leftarrow\mathsf{Setup}(1^\lambda)$*,* $vk\leftarrow\mathsf{KeyGen}(pk,\tau)$*, and every* (σ_f,y,f) *with* $\mathsf{Verify}(vk,\tau,\sigma_f,f)=1$*,* $\Delta\big(\mathsf{Hide}(vk,y,\sigma_f),\mathsf{Sim}(sk,vk,\tau,y,f)\big)\leq\mathsf{negl}(\lambda)$*, where* $\Delta(\cdot,\cdot)$ *denotes statistical distance, following from the* SampleLeft–SampleRight *statistical closeness of [26].*

For any valid $\sigma_f=(\mathbf{R}_f,y,f)$, Hide calls $\mathsf{SampleRight}(\mathbf{H}_f,\mathbf{R}_f,\mathbf{z})$ with $\mathbf{H}_f=[\mathbf{A}_\tau \mid \mathbf{D}_f+(y-1)\mathbf{G}_n]$, while $\mathsf{Sim}(sk,vk,\tau,y,f)$ recomputes the same $\mathbf{H}_f$ and calls SampleLeft from the trapdoor without accessing $\mathbf{R}_f$. The SampleLeft–SampleRight statistical closeness of [26] implies $\Delta(\mathsf{Hide}(vk,y,\sigma_f),\mathsf{Sim}(sk,vk,\tau,y,f))\leq\mathsf{negl}(\lambda)$, so $\tilde{\sigma}_f$ reveals nothing beyond y.

4.2 Efficiency Comparison with Gurbanov et al. [26]

Here, we discuss how our scheme's time and space cost is more efficient than Gurbanov et al. [26], even by adding context-hiding property.

Signing complexity. In [26], Sign invokes SamPre once per data element $i\in[N]$, costing $O(N\cdot m^2n)$ time and $O(N\cdot m^2\log q)$ bits of storage. Our FHS calls SampleLeft exactly *twice* (for $k\in\{0,1\}$), regardless of ℓ; each per-message signature is then derived via the deterministic multiplication $[\mathbf{R}^\top_{\mu_i,1,i}\,|\,\mathbf{R}^\top_{\mu_i,2,i}]^\top=[\mathbf{R}^\top_{k,1}\,|\,\mathbf{R}^\top_{k,2}]^\top\cdot\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n)$ at $O(m^2)$ per message (since $\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n)\in\{0,1\}^{m\times m}$), giving total cost $O((\ell+n)\cdot m^2)$—a factor-$\Theta(n)$ improvement for $\ell\gg1$.

$\mathsf{Exp}^{\mathsf{du\text{-}mb\text{-}BPRIV},\beta}_{\mathcal{A},\mathcal{V},\mathsf{Sim},\mathsf{Recover}}(\lambda)$

1: $Checked, Happy \leftarrow \emptyset; V, PU, U, CU \leftarrow \mathsf{empty}$
2: $(pd, sd) \leftarrow \mathsf{Setup}(1^\lambda)$
3: **for** $id \in \mathcal{I}$ **do**
4: $\quad (pc, sc) \leftarrow \mathsf{Register}(id); U[id] \leftarrow sc; PU[id] \leftarrow pc$
5: $\quad$ **if** $id \in \mathcal{D}$ **then** $CU[id] \leftarrow U[id]$
6: $BB \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{voteLR}}, \mathcal{O}_{\mathsf{board}}}(pd, CU, PU, \mathcal{H}_{\mathsf{check}})$
7: **if** $\mathcal{H}_{\mathsf{check}} \not\subseteq V \vee \mathsf{ValidBoard}(BB, pd) = \perp$ **then return** $d \leftarrow\$ \{0,1\}$
8: $d^* \leftarrow \mathcal{A}()$
9: $(r^*, \Pi^*) \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{tally}}}()$
10: **if** $\mathsf{VerifyTally}((pd, pbb, r^*), \Pi^*) = \perp$ **then return** $d \leftarrow\$ \{0,1\}$
11: $d \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{verify}}}()$
12: **if** $\mathcal{H}_{\mathsf{check}} \not\subseteq Checked$ **then return** $d \leftarrow\$ \{0,1\}$
13: **if** $\mathcal{H}_{\mathsf{check}} \not\subseteq Happy$ **then return** d^*
14: **return** d

$\mathcal{O}_{\mathsf{board}}()$

1: **return** $\mathsf{Publish}(BB_\beta)$

$\mathcal{O}_{\mathsf{voteLR}}(id, v_0, v_1)$

1: **if** $id \notin \mathcal{H}$ **then return** $\perp$
2: **for** $i \in \{0,1\}$ **do** $(pc, b_i, st^i_{\mathsf{pre}}, st^i_{\mathsf{post}}) \leftarrow \mathsf{Vote}(pd, pc, v_i)$
3: $V[id] \leftarrow V[id] \| (st^\beta_{\mathsf{pre}}, st^0_{\mathsf{post}}, v_0)$
4: $BB_0 \leftarrow BB_0 \| (id, (pc, b_0)); BB_1 \leftarrow BB_1 \| (id, (pc, b_1))$

$\mathcal{O}_{\mathsf{tally}}()$

1: **if** $\beta = 0$ **then** $(r, \Pi) \leftarrow \mathsf{Tally}(BB, pd, sd)$; **return** (r, Π)
2: $BB' \leftarrow \mathsf{Recover}(BB, BB_0, BB_1)$
3: $(r, \Pi) \leftarrow \mathsf{Tally}(BB', pd, sd)$
4: **return** $(r, \mathsf{Sim}(pd, \mathsf{Publish}(BB), r))$

$\mathcal{O}_{\mathsf{verify}}(id)$ for $id \in \mathcal{H}_{\mathsf{check}}$

1: $Checked \leftarrow Checked \cup \{id\}$
2: **if** $\mathsf{VerifyVote}(id, st_{\mathsf{pre}}, st_{\mathsf{post}}, BB, pc, sc) = \top$ **then**
3: $\quad Happy \leftarrow Happy \cup \{id\}$
4: **return** $Happy$

Fig. 3: The du-mb-BPRIV security experiment. $\mathcal{I} = \mathcal{H} \cup \mathcal{D}$ with $\mathcal{H}_{\mathsf{check}} \subseteq \mathcal{H}$.

Context-hiding overhead and space cost. Both schemes achieve statistical context hiding via a single SampleRight/SampleLeft call, adding $\widetilde{O}(m^2 n)$ overhead. Our hiding matrix $\mathbf{H}_f = [\mathbf{A}_\tau \mid \mathbf{D}_f + (y-1)\mathbf{G}_n] \in \mathbb{Z}_q^{n \times 3m}$ yields $\tilde{\sigma}_f \in \mathbb{Z}^{3m}$ versus $\tilde{\sigma} \in \mathbb{Z}^{2m}$ in [26]—a modest constant-factor increase. For public-key size, [26] requires $O(N \cdot nm\log q)$ bits for $(\mathbf{V}_1, ..., \mathbf{V}_N)$, while our $pk = (\mathbf{A}, \mathbf{A}^*, \mathbf{D}_0, \mathbf{D}_1, \mathbf{B}, \mathbf{B}', \mathbf{z})$ occupies only $O(nm\log q)$ bits independent of ℓ, with $\mathbf{z} \in \mathbb{Z}_q^n$ adding a negligible $O(n\log q)$ bits. Secret keys are identical in both $(O(m^2 \log q))$. Overall, our FHS achieves two pre-image samples instead of ℓ, dataset-independent public keys, and context hiding—at the cost of only one extra element $\mathbf{z}$ and a hidden signature of size $3m$ versus $2m$.

5 Security Analysis of Phoenix

Ballot Privacy. We consider the strong correctness, strong consistency, and ideal functionality from [9]. The most fundamental voting privacy property that has been well-studied is BPRIV [9] (no entity can learn the content of ballots), and an stronger notion is mb-BPRIV [93] (ballot privacy against a adversary who controls the ballot box). du-mb-BPRIV (mb-BPRIV where verification is delayed until after tallying) [28], which we focus on it here, is equivalent (or in some sense slightly stronger than mb-BPRIV [28,93]).

Definition 11 (du-mb-BPRIV [28]). *Let $\mathcal{V}$ be a voting system,* Recover *a recovery algorithm, and $\mathcal{I} = \mathcal{H} \cup \mathcal{D}$ a partition into honest and dishonest voters with $\mathcal{H}_{\mathsf{check}} \subseteq \mathcal{H}$ the verifying subset. A* selection function *for integers $m, n \geq 1$ is a mapping $\pi : [n] \to [m] \cup (\{0,1\}^* \times \{0,1\}^*)$, inducing a board transformation $\bar{\pi}$; the recovery algorithm* $\mathsf{Recover}(BB, BB_1)$ *returns such a π capturing how $\mathcal{A}$ constructed BB from BB_1. We say $\mathcal{V}$ satisfies* du-mb-BPRIV *w.r.t.* Recover *if there exists an efficient simulator* Sim *such that for every efficient adversary $\mathcal{A}$, $\mathsf{Adv}^{\mathit{du\text{-}mb\text{-}bpriv}}_{\mathcal{A},\mathcal{V},\mathsf{Sim}}(\lambda) = \left| \Pr[\mathsf{Exp}^0 = 1] - \Pr[\mathsf{Exp}^1 = 1] \right| = \mathsf{negl}(\lambda)$, where $\mathsf{Exp}^\beta := \mathsf{Exp}^{\mathit{du\text{-}mb\text{-}BPRIV},\beta}_{\mathcal{A},\mathcal{V},\mathsf{Sim},\mathsf{Recover}}(\lambda)$ is defined in Figure 3.*

In the experiment, $\mathcal{O}_{\mathsf{voteLR}}$ computes ballots for both v_0,v_1, storing them in BB_0,BB_1; the voter state records $st^{\beta}_{\mathsf{pre}}$ (dependent on the secret bit) and st^{0}_{post} (always from $\beta=0$, since recovery ensures the tally reflects the left board). After $\mathcal{A}$ outputs board BB, it makes an initial guess d^* *before* seeing the tally. $\mathcal{O}_{\mathsf{tally}}$ (single call): for $\beta=0$ tallies BB directly; for $\beta=1$ recovers $BB' \leftarrow \mathsf{Recover}(BB,BB_0,BB_1)$, tallies BB', and returns the result with a simulated proof. $\mathcal{O}_{\mathsf{verify}}$ runs $\mathsf{VerifyVote}$ for voters in $\mathcal{H}_{\mathsf{check}}$. If any expected verification is missing or fails, the experiment returns d^* (or a random bit), ensuring the adversary gains no advantage from tampering that causes detectable complaints.

Theorem 8. *Let* BFV *be IND-CPA secure under* $\mathsf{RLWE}_{n,q,\chi}$*,* ZKP *be computationally zero-knowledge and knowledge-sound under* MSIS *and* MLWE*,* FHS *be existentially unforgeable under* $\mathsf{SIS}_{n,m,q,\beta}$*, and* H *be collision-resistant. Then for every PPT adversary* $\mathcal{A}$ *making at most* q_v *queries to* $\mathcal{O}_{\mathsf{voteLR}}$*, there exist a simulator* Sim*, a recovery algorithm* $\mathsf{Recover}$*, and PPT reductions* $\mathcal{B}_{\mathsf{cpa}}$*,* $\mathcal{B}_{\mathsf{zk}}$*,* $\mathcal{B}_{\mathsf{fhs}}$*,* $\mathcal{B}_{\mathsf{cr}}$ *such that*
$\mathsf{Adv}^{\textit{du-mb-bpriv}}_{\mathcal{A},\mathsf{Phoenix},\mathsf{Sim}}(\lambda) \leq q_v \cdot \mathsf{Adv}^{\mathsf{ind\text{-}cpa}}_{\mathcal{B}_{\mathsf{cpa}},\mathsf{BFV}}(\lambda) + q_v \cdot \mathsf{Adv}^{\mathsf{zk}}_{\mathcal{B}_{\mathsf{zk}},\mathsf{ZKP}}(\lambda) + \mathsf{Adv}^{\mathsf{euf\text{-}acma}}_{\mathcal{B}_{\mathsf{fhs}},\mathsf{FHS}}(\lambda) + \mathsf{Adv}^{\mathsf{cr}}_{\mathcal{B}_{\mathsf{cr}},H}(\lambda) + \mathsf{negl}(\lambda)$.

Intuitively, $\mathsf{Phoenix}$ achieves $\mathsf{du\text{-}mb\text{-}BPRIV}$ because IND-CPA security of BFV renders the ciphertexts on the adversary's view of the board independent of the real votes, computational zero-knowledge of ZKP extends this indistinguishability to the accompanying well-formedness proofs (and hence, by composition, to the FHS signatures computed over them), existential unforgeability of FHS prevents the adversary from substituting or injecting ballots under honest credentials, thereby making sure that the recovery algorithm correctly reconstructs a board consistent with the left-world votes for tallying, and collision resistance of H guarantees that the hash-based individual verification check cannot be exploited to distinguish the two worlds.

Strong Consistency. This property demands that for any (adversarially produced) bulletin board for which each individual ballot is valid, the result provided by the Tally algorithm is the correct one. The original definition associated voter identities with ballots; here, we use their public credentials instead.

Definition 12. *A voting scheme is* strongly consistent *if there exists:*

- *An extraction algorithm* $\mathsf{Extract}(b,\mathsf{sk})$ *that provides the upk with the value computed by decrypting of the ciphertext contained in the ballot* b *with the secret sk; and*
- *A ballot validation algorithm* $\mathsf{ValidInd}(b,\mathsf{pk})$ *returns true iff ballot* b *is "well-formed" with respect to some notion of well-formedness determined in advance by election.*

These algorithms must satisfy the following conditions:

- *For any* $(\mathsf{pk},\mathsf{sk})$ *obtained from* $\mathsf{Setup}(\lambda)$*, and any* (id,v,usk) *with* upk *trivially computed from* usk*, if* $b \leftarrow \mathsf{Vote}(id,v,\mathsf{pk},\mathsf{usk})$ *then* $\mathsf{Extract}(b,\mathsf{sk})$ *returns* (upk,v) *with overwhelming probability.*
- *For any adversarially produced* (BB, b)*, if* $\mathsf{Valid}(\mathsf{BB}, b, \mathsf{pk})$ *returns true, then* $\mathsf{ValidInd}(b,\mathsf{pk})$ *returns true as well.*
- *For any adversary* $\mathcal{B}$ *that returns a ballot box with ballots that satisfy* $\mathsf{ValidInd}$*, the experiment* $\mathsf{Exp}^{\mathsf{consis}}_{\mathcal{B},\mathcal{V},I}(\lambda)$ *specified in Figure 4 returns true with a probability negligible in the security parameter.*

$\mathsf{Exp}^{\mathsf{consis}}_{\mathcal{B},\mathcal{V},I}(\lambda)$

```
uL ← empty
(pk,sk) ← Setup(1^λ)
∀id ∈ I do uL.[id] ← Register(id)
BB ← B(pk,uL)
e ← ∀b ∈ BB. ValidInd(b,pk)
(r,Π) ← Tally(BB,sk)
for i in 1..|BB| do
    dbb[i] ← Extract(BB[i],sk)
r' ← Count∘Policy(dbb)
return (r ≠ r' ∧ e)
```

$\mathsf{Exp}^{\mathsf{corr}}_{\mathcal{B},\mathcal{V},I}(\lambda)$

```
uL ← empty
(pk,sk) ← Setup(1^λ)
∀id ∈ I do uL.[id] ← Register(id)
(id,v,BB) ← B(pk,uL)
e ← false
if (id ∈ I) then
    (usk,upk) ← uL.[id]
    e1 ← ∀x.∀y. (x,y,⋆,⋆) ∈ BB ⟹
        (x,y) = (id,upk) ∨ (x ≠ id ∧ y ≠ upk)
    b ← Vote(id,v,pk,usk)
    e2 ← Valid(BB,b,pk)
    e ← e1 ∧ ¬e2
return e
```

Fig. 4: Security experiments for strong consistency (left, $\mathsf{Exp}^{\mathsf{consis}}$) and strong correctness (right, $\mathsf{Exp}^{\mathsf{corr}}$). The adversary $\mathcal{B}$ wins $\mathsf{Exp}^{\mathsf{consis}}$ if tallying produces a result inconsistent with the extracted ballots despite all ballots being individually valid. The adversary wins $\mathsf{Exp}^{\mathsf{corr}}$ if it can submit a ballot bulletin board that renders an honest voter's fresh ballot invalid.

Theorem 9. *Let* BFV *be correct and IND-CPA-secure under* $\mathsf{RLWE}_{n,q,\chi}$*, let* ZKP *be complete and knowledge-sound under* MSIS*, and let* FHS *be existentially unforgeable under* $\mathsf{SIS}_{n,m,q,\beta}$*. Define* $\mathsf{Extract}(b,sk)$ *to parse* $b=(id,upk,c,\sigma,s)$ *and return* $(upk,\mathsf{BFV.Dec}(sk,c))$*, and* $\mathsf{ValidInd}(b,pk)$ *to return* 1 *iff* $\mathsf{ZKP.Verify}(pp,(c,upk),\sigma)=1$ *and* $\mathsf{FHS.Verify}(upk,id,(c,\sigma),s)=1$*. Then for every PPT adversary* $\mathcal{B}$*, we have* $\Pr\left[\mathsf{Exp}^{\mathsf{consis}}_{\mathcal{B},\mathsf{Phoenix},\mathcal{I}}(\lambda)=1\right]\leq\mathsf{Adv}^{\mathsf{ks}}_{\mathsf{ZKP}}(\lambda)+\mathsf{negl}(\lambda)$*, where the negligible term accounts for the probability that* BFV *decryption fails on a correctly encrypted plaintext.*

In Phoenix, if every $b\in BB$ satisfies ValidInd, knowledge soundness of ZKP guarantees each c encrypts a valid $v\in\{0,1\}$, and unforgeability of FHS binds each (c,σ) to its registered upk. Correctness of BFV then makes sure Extract recovers (upk,v), and additive homomorphism yields $r=\sum v_i=\mathsf{Count}(dbb)$ with overwhelming probability.

Strong Correctness. This is the guarantee that an honestly created ballot will not be rejected by a ballot validation algorithm.

Definition 13 (Strong Correctness). *A voting scheme* $\mathcal{V}$ *is* strongly correct *if the advantage of any efficient adversary* $\mathcal{B}$*, defined by* $\mathcal{A}^{\mathsf{corr}}_{\mathcal{B},\mathcal{V},I}(\lambda)=\Pr[\mathsf{Exp}^{\mathsf{corr}}_{\mathcal{B},\mathcal{V},I}(\lambda)=1]$ *(where* $\mathsf{Exp}^{\mathsf{corr}}_{\mathcal{B},\mathcal{V},I}(\lambda)$ *is defined in Figure 4) is negligible as a function of* λ*.*

Theorem 10. *Let* BFV *be correct, let* ZKP *be complete, and let* FHS *be correct (i.e.* FHS.Verify *accepts honestly generated signatures with overwhelming probability). Then for every PPT adversary* $\mathcal{B}$*,* $\mathsf{Adv}^{\mathsf{corr}}_{\mathcal{B},\mathsf{Phoenix},\mathcal{I}}(\lambda)=\Pr\left[\mathsf{Exp}^{\mathsf{corr}}_{\mathcal{B},\mathsf{Phoenix},\mathcal{I}}(\lambda)=1\right]\leq\mathsf{negl}(\lambda)$*.*

In Phoenix, an honest ballot runs BFV.Enc, ZKP.Prove, and FHS.Sign with correctly generated keys, so completeness and correctness of the respective primitives ensure $e_2=e_3=1$. The credential-consistency check e_1 holds because the experiment enforces no conflicting (id,upk) pair unless $\mathcal{B}$ itself placed it. Hence $\mathsf{Valid}(BB,b,pk)=1$ with overwhelming probability.

Ideal Functionality. This property states the voting protocol execution is indistinguishable from interaction with a trusted party.

Definition 14 (Ideal Voting Functionality). *Let $\mathcal{I}$ be a set of voter identities, $\mathbb{V}$ a set of possible votes, and $\rho:(\mathcal{I}\times\mathbb{V})^*\to\mathcal{R}$ a result function mapping sequences of identity-vote pairs to a result space $\mathcal{R}$. The ideal functionality $\mathcal{F}_{\text{voting}}(\rho)$, interacting with an environment $\mathfrak{E}$ (controlling honest voters) and an ideal-world adversary (simulator) $\mathcal{S}$ (acting on behalf of corrupted voters), maintains an initially empty list L of identity-vote pairs and proceeds as follows:*

1. ***Vote:*** *On input* $\mathsf{vote}(id,v)$ *from $\mathfrak{E}$ or $\mathcal{S}$, append (id,v) to L and send* $\mathsf{ack}(id)$ *to $\mathcal{S}$.*
2. ***Tally:*** *On input* tally *from $\mathcal{S}$, compute $r\leftarrow\rho(L)$ and return r to both $\mathfrak{E}$ and $\mathcal{S}$, then halt.*

The full ideal-world execution, in which $\mathfrak{E}$, $\mathcal{S}$, and $\mathcal{F}_{\text{voting}}(\rho)$ interact and $\mathfrak{E}$ outputs a single bit, is denoted $\beta\leftarrow\mathsf{idealexec}(\mathfrak{E}\|\mathcal{S}\|\mathcal{F}_{\text{voting}}(\rho))$.

Theorem 11. *If* Phoenix *is du-mb-BPRIV (Theorem 8), strongly consistent (Theorem 9), and strongly correct (Theorem 10), then for every PPT environment $\mathfrak{E}$ there exists a PPT simulator $\mathcal{S}$ such that*

$$\left|\Pr[\mathsf{realexec}(\mathfrak{E}\|\mathcal{A}\|\mathsf{Phoenix})=1]-\Pr[\mathsf{idealexec}(\mathfrak{E}\|\mathcal{S}\|\mathcal{F}_{\text{voting}}(\rho))=1]\right|\le\mathsf{negl}(\lambda).$$

By the composition theorem of Cortier et al. [9], any scheme satisfying mb-BPRIV, strong correctness, and strong consistency realises $\mathcal{F}_{\text{voting}}(\rho)$: strong correctness lets $\mathcal{S}$ simulate the honest voting interface; strong consistency ensures the real-world tally matches ρ on extracted votes; and mb-BPRIV lets $\mathcal{S}$ simulate the adversary's view from r alone. Since du-mb-BPRIV implies mb-BPRIV and Phoenix satisfies all three properties by Theorems 8–10, the composition applies directly.

Verifiability. Generally, it comes from the fact that voters can verify their ballots were recorded and anyone can verify correct tallying, but we consider the *Strong Verifiability* notion by [10] here, which covers broader spectrum.

Definition 15 (Strong Verifiability). *Let $\mathcal{V}=$* (Setup, Credential, Vote, Validate, Box, VerifyVote, Tally, Verify) *be a voting protocol with admissible vote space $\mathbb{V}$, result space $\mathcal{R}$, and result function $\rho:\mathbb{V}^\tau\to\mathcal{R}$ admitting partial tallying via a commutative operation $\oplus_R$, i.e. $\rho(S_1\cup S_2)=\rho(S_1)\oplus_R\rho(S_2)$ for disjoint vote lists $S_1,S_2\subseteq\mathbb{V}^*$. Both experiments maintain: a public credential set L; a table U of registered tuples (id,upk,usk); a set CU of corrupted voter pairs (id,upk); a list $HVote$ of triples (id,v,b) from honest voting queries (last ballot per id only); and $Checked\subseteq HVote$ where* $\mathsf{CheckedVoter}(id,v,b)=1$. *The oracles are: $\mathcal{O}_{\mathsf{reg}}(id)$: honestly generates $(upk,usk)\leftarrow\mathsf{Credential}(1^\lambda,id)$, returns upk, updates L,U. $\mathcal{O}_{\mathsf{corrupt}}(id)$: reveals (upk,usk) for registered id, updates CU. $\mathcal{O}_{\mathsf{vote}}(id,v)$: for honest, uncorrupted id with $v\in\mathbb{V}$, returns $b\leftarrow\mathsf{Vote}(id,upk,usk,v)$, updates $HVote$. $\mathcal{O}_{\mathsf{cast}}(id,b)$: runs $\mathsf{Box}(BB,b)$ on an honest bulletin board.*

The experiments are given in Figure 5. In both, the adversary wins if it produces a verified tally that cannot be expressed as $\rho(\{v_i^E\})\oplus_R\rho(\{v_i^A\})\oplus_R\rho(\{v_i^B\})$, where $\{v_i^E\}$ are votes from Checked, $\{v_i^A\}$ are from some subset of $HVote\setminus Checked$, and $\{v_i^B\}\subseteq\mathbb{V}$ with $|\{v_i^B\}|\le|CU|$. We say $\mathcal{V}$ achieves strong verifiability *if for all PPT $\mathcal{A}$, both $\Pr[\mathsf{Exp}^{\mathsf{ver\text{-}b}}_{\mathcal{A},\mathcal{V}}(\lambda)=1]$ and $\Pr[\mathsf{Exp}^{\mathsf{ver\text{-}r}}_{\mathcal{A},\mathcal{V}}(\lambda)=1]$ are negligible, i.e. under the assumption that the registration authority and bulletin board are not simultaneously dishonest.*

$\mathsf{Exp}^{\mathsf{ver\text{-}b}}_{\mathcal{A},\mathcal{V}}(\lambda)$

$L,U,CU,HVote \leftarrow \emptyset$
$(pk,sk) \leftarrow \mathsf{Setup}(1^\lambda)$
$(BB,\rho,\Pi) \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{reg}},\mathcal{O}_{\mathsf{corrupt}},\mathcal{O}_{\mathsf{vote}}}(pk)$
if $\mathsf{Verify}(BB,\rho,\Pi)=0 \vee \rho=\perp$ **then return** 0
if $\exists\{(id_i^A,v_i^A,\star)\} \subseteq HVote \setminus Checked, \exists\{v_i^B\} \subseteq \mathbb{V}, |\{v_i^B\}| \leq |CU|$:
  $\rho=\rho(\{v_i^E\}) \oplus_R \rho(\{v_i^A\}) \oplus_R \rho(\{v_i^B\})$ **then return** 0
return 1

$\mathsf{Exp}^{\mathsf{ver\text{-}r}}_{\mathcal{A},\mathcal{V}}(\lambda)$

$L,U,CU,HVote \leftarrow \emptyset$
$(pk,sk) \leftarrow \mathsf{Setup}(1^\lambda)$
$(\rho,\Pi) \leftarrow \mathcal{A}^{\mathcal{O}_{\mathsf{vote}},\mathcal{O}_{\mathsf{corrupt}},\mathcal{O}_{\mathsf{cast}}}(pk)$
if $\mathsf{Verify}(BB,\rho,\Pi)=0 \vee \rho=\perp$ **then return** 0
if $\exists\{(id_i^A,v_i^A,\star)\} \subseteq HVote \setminus Checked, \exists\{v_i^B\} \subseteq \mathbb{V}, |\{v_i^B\}| \leq |CU|$:
  $\rho=\rho(\{v_i^E\}) \oplus_R \rho(\{v_i^A\}) \oplus_R \rho(\{v_i^B\})$ **then return** 0
return 1

Fig. 5: $\mathsf{Exp}^{\mathsf{ver\text{-}b}}$ (left): $\mathcal{A}$ controls trustee, voters, and bulletin board; registrar is honest. $\mathsf{Exp}^{\mathsf{ver\text{-}r}}$ (right): $\mathcal{A}$ controls trustee, voters, and registrar; bulletin board is honest ($\mathcal{A}$ adds ballots only via $\mathcal{O}_{\mathsf{cast}}$, does not output BB).

Theorem 12. *Let H be collision-resistant, let* ZKP *be knowledge-sound under* MSIS*, let* FHS *be existentially unforgeable under* $\mathsf{SIS}_{n,m,q,\beta}$*, and let* BFV *be correct. Then for every PPT adversary $\mathcal{A}$,* $\Pr\left[\mathsf{Exp}^{\mathsf{ver\text{-}b}}_{\mathcal{A},\mathsf{Phoenix}}(\lambda)=1\right] \leq \mathsf{Adv}^{\mathsf{euf}}_{\mathsf{FHS}}(\lambda)+\mathsf{Adv}^{\mathsf{ks}}_{\mathsf{ZKP}}(\lambda)+\mathsf{Adv}^{\mathsf{cr}}_{H}(\lambda)+\mathsf{negl}(\lambda)$*, and likewise* $\Pr[\mathsf{Exp}^{\mathsf{ver\text{-}r}}_{\mathcal{A},\mathsf{Phoenix}}(\lambda)=1] \leq \mathsf{negl}(\lambda)$ *under the same assumptions.*

Individual verifiability follows from collision resistance of H and unforgeability of FHS: substituting or omitting a ballot requires a hash collision or a signature forgery under the voter's *upk*. Universal verifiability follows from knowledge soundness of ZKP: an accepting tally proof guarantees r is the correct aggregate. Eligibility verifiability follows from unforgeability of FHS: every accepted ballot carries a valid signature under a registered *upk*, bounding adversarial contributions to $|CU|$.

6 Future Directions

An FHS is used in Phoenix instead of a classical signature scheme because unlike other post-quantum schemes [4, 14–23], we plan to deploy an actual production-ready implementation of Phoenix as an online web-application voting platform, which in next update, the context-hiding property of FHS [94], along with using fake credentials [95], will get used to achieve stronger privacy guarantees, such as receipt-freeness [96] (voters cannot construct proofs of their choices to third parties) and coercion-resistance [97] (the coercer cannot know whether voter followed his instruction or not).

We aim to eliminate the distributed key generation phase on real-world deployment of Phoenix via silent-setup threshold cryptosystems [98]; however, current constructions rely on witness encryption [99], which remains inefficient and quantum-vulnerable [100, 101]. Designing a post-quantum *silent-setup* threshold public key scheme that can support homomorphic operations remains an open problem. In addition, except using signatures on randomizable ciphertexts [102], no solution has been proposed to achieve strong receipt-freeness [96], so using the same technique on PQ voting can be progressive, but designing post-quantum signatures on randomizable ciphertexts remains as an open problem.

A Appendix

A.1 Proof of Theorem 1

Proof. Let $(pk, sk) \leftarrow \mathsf{Setup}(1^\lambda, 1^{|\tau|}, 1^\ell, 1^{d_{\max}})$ and $vk \leftarrow \mathsf{KeyGen}(pk, \tau)$. For each $k \in \{0,1\}$, the $\mathsf{SampleLeft}$ algorithm produces a short matrix $[\mathbf{R}_{k,1}^\top \,|\, \mathbf{R}_{k,2}^\top]^\top$ satisfying

$$\mathbf{A}_\tau \begin{bmatrix} \mathbf{R}_{k,1} \\ \mathbf{R}_{k,2} \end{bmatrix} = \mathbf{D}_k + [k\mathbf{I}_n \,|\, \cdots \,|\, k\mathbf{I}_n]\mathbf{G}_n. \tag{1}$$

For a message $\mu_i \in \{0,1\}$ at index $i \in [\ell]$, setting $k = \mu_i$, the Unpack procedure computes

$$\begin{bmatrix} \mathbf{R}_{\mu_i,1,i} \\ \mathbf{R}_{\mu_i,2,i} \end{bmatrix} = \begin{bmatrix} \mathbf{R}_{k,1} \\ \mathbf{R}_{k,2} \end{bmatrix} \cdot \mathbf{G}_n^{-1}(\mathbf{P}_i \mathbf{G}_n).$$

Right-multiplying both sides of (1) by $\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n)$ yields

$$\begin{aligned} \mathbf{A}_\tau \begin{bmatrix} \mathbf{R}_{\mu_i,1,i} \\ \mathbf{R}_{\mu_i,2,i} \end{bmatrix} &= \big(\mathbf{D}_k + [k\mathbf{I}_n \,|\, \cdots \,|\, k\mathbf{I}_n]\mathbf{G}_n\big) \cdot \mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n) \\ &= \mathbf{D}_{\mu_i}\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n) + [k\mathbf{I}_n \,|\, \cdots \,|\, k\mathbf{I}_n]\mathbf{G}_n\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n) \\ &= \mathbf{D}_{\mu_i}\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n) + [k\mathbf{I}_n \,|\, \cdots \,|\, k\mathbf{I}_n]\mathbf{P}_i\mathbf{G}_n \\ &= \mathbf{D}_{\mu_i}\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n) + \mu_i\mathbf{G}_n, \end{aligned} \tag{2}$$

where the last equality uses the property of the extended unit matrix $\mathbf{P}_i = [\mathbf{0}_n \,|\, \cdots \,|\, \mathbf{I}_n \,|\, \cdots \,|\, \mathbf{0}_n]$, which extracts the i-th block so that $[k\mathbf{I}_n \,|\, \cdots \,|\, k\mathbf{I}_n]\mathbf{P}_i = k\mathbf{I}_n$ and hence $[k\mathbf{I}_n \,|\, \cdots \,|\, k\mathbf{I}_n]\mathbf{P}_i\mathbf{G}_n = k\mathbf{G}_n = \mu_i\mathbf{G}_n$.

The public matrix associated with the identity circuit id_i is $\mathbf{D}_{\mathsf{id}_i} = \mathbf{D}_{\mu_i}\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n)$. Since $\mathsf{SampleLeft}$ makes sure $\|[\mathbf{R}_{k,1}^\top \,|\, \mathbf{R}_{k,2}^\top]^\top\|_\infty \le s\sqrt{m}$ and $\mathbf{G}_n^{-1}(\mathbf{P}_i\mathbf{G}_n) \in \{0,1\}^{m\times m}$, we have $\|[\mathbf{R}_{\mu_i,1,i}^\top \,|\, \mathbf{R}_{\mu_i,2,i}^\top]^\top\|_\infty \le m \cdot s\sqrt{m} \le B$ for the chosen parameters. Together with (2), the verification check $\mathbf{A}_\tau\mathbf{R}_{\sigma_i} = \mathbf{D}_{\mathsf{id}_i} + \mu_i\mathbf{G}_n$ and $\|\mathbf{R}_{\sigma_i}\|_\infty \le B$ both hold, so $\mathsf{Verify}(vk, \tau, \sigma_i, \mathsf{id}_i) = 1$. □

A.2 Proof of Theorem 2

Proof. We proceed by induction on the circuit structure. Since every Boolean circuit can be expressed using NAND gates, it suffices to verify correctness for a single NAND gate $g = (u, v, w)$ with inputs $x, y \in \{0,1\}$.

Suppose that at the input wires of gate g, we have signatures $\mathbf{R}_u, \mathbf{R}_v \in \mathbb{Z}^{2m\times m}$ satisfying

$$\mathbf{A}_\tau\mathbf{R}_u = \mathbf{D}_u\mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n) + x\mathbf{G}_n \quad \text{and} \quad \mathbf{A}_\tau\mathbf{R}_v = \mathbf{D}_v\mathbf{G}_n^{-1}(\mathbf{P}_{y_j}\mathbf{G}_n) + y\mathbf{G}_n.$$

The base case holds by signing correctness (Theorem 1).

The Eval algorithm computes $\hat{\mathbf{D}}_u = \mathbf{G}_n^{-1}(\mathbf{D}_u\mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n)) \in \{0,1\}^{m\times m}$, the output signature $\mathbf{R}_w = \mathbf{R}_v\hat{\mathbf{D}}_u - y\mathbf{R}_u$, and the public matrix $\mathbf{D}_w = \mathbf{D}_v\mathbf{G}_n^{-1}(\mathbf{P}_{y_j}\mathbf{G}_n)\hat{\mathbf{D}}_u - \mathbf{G}_n$.

We verify:

$$\begin{aligned}
\mathbf{A}_\tau \mathbf{R}_w &= \mathbf{A}_\tau(\mathbf{R}_v \hat{\mathbf{D}}_u - y\mathbf{R}_u) \\
&= (\mathbf{A}_\tau \mathbf{R}_v)\hat{\mathbf{D}}_u - y(\mathbf{A}_\tau \mathbf{R}_u) \\
&= \big(\mathbf{D}_v \mathbf{G}_n^{-1}(\mathbf{P}_{y_j}\mathbf{G}_n) + y\mathbf{G}_n\big)\hat{\mathbf{D}}_u - y\big(\mathbf{D}_u \mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n) + x\mathbf{G}_n\big) \\
&= \mathbf{D}_v \mathbf{G}_n^{-1}(\mathbf{P}_{y_j}\mathbf{G}_n)\hat{\mathbf{D}}_u + y\mathbf{G}_n\hat{\mathbf{D}}_u - y\mathbf{D}_u \mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n) - xy\mathbf{G}_n \\
&= \mathbf{D}_v \mathbf{G}_n^{-1}(\mathbf{P}_{y_j}\mathbf{G}_n)\hat{\mathbf{D}}_u + y\mathbf{D}_u \mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n) - y\mathbf{D}_u \mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n) - xy\mathbf{G}_n \\
&= \mathbf{D}_v \mathbf{G}_n^{-1}(\mathbf{P}_{y_j}\mathbf{G}_n)\hat{\mathbf{D}}_u - xy\mathbf{G}_n \\
&= \mathbf{D}_w + \mathbf{G}_n - xy\mathbf{G}_n \\
&= \mathbf{D}_w + (1-xy)\mathbf{G}_n = \mathbf{D}_w + (x\,\mathsf{NAND}\,y)\mathbf{G}_n,
\end{aligned}$$

where in the fifth equality we used $\mathbf{G}_n\hat{\mathbf{D}}_u = \mathbf{G}_n\mathbf{G}_n^{-1}(\mathbf{D}_u\mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n)) = \mathbf{D}_u\mathbf{G}_n^{-1}(\mathbf{P}_{x_i}\mathbf{G}_n)$.

For the noise bound, since $\hat{\mathbf{D}}_u \in \{0,1\}^{m\times m}$ and $|y| \le 1$, we have $\|\mathbf{R}_w\|_\infty \le m\|\mathbf{R}_v\|_\infty + \|\mathbf{R}_u\|_\infty$. By induction over a circuit of depth $d_{\max}$, the noise grows as $\|\mathbf{R}_f\|_\infty \le (m+1)^{d_{\max}} \cdot \beta_{\text{init}} \le B$ for appropriately chosen parameters.

Composed evaluation follows by the same inductive argument: if σ_j certifies $h_j(\boldsymbol{\mu})$ under public matrix $\mathbf{D}_{h_j}$, then evaluating g on $(\sigma_1, \dots, \sigma_s)$ yields σ^* certifying $g(h_1(\boldsymbol{\mu}),\dots,h_s(\boldsymbol{\mu}))$ under $\mathbf{D}_{g\circ\bar{h}}$, provided the total depth does not exceed $d_{\max}$. □

A.3 Proof of Theorem 3

Proof. The evaluated signature $\sigma_f = (\mathbf{R}_f, f(\boldsymbol{\mu}), f)$ has its dominant component $\mathbf{R}_f \in \mathbb{Z}_q^{2m\times m}$. Each entry of $\mathbf{R}_f$ can be represented with $\lceil \log q \rceil$ bits, so $|\mathbf{R}_f| = 2m^2\lceil \log q \rceil$. Recall $m = O(n \cdot d_{\max})$ and $q = n^{O(d_{\max})}$ for polylogarithmic-depth circuits, so $\lceil \log q \rceil = O(d_{\max}\log n)$ and $|\mathbf{R}_f| = \text{poly}(\lambda, d_{\max})$.

The representation size of $\mathbf{R}_f$ is *independent* of the circuit size $|f|$ and of the number of inputs ℓ. The dataset size ℓ enters only through the vector encoding's use of extended unit matrices $\mathbf{P}_i \in \mathbb{Z}^{\ell n\times n}$; however, the dimension of the gadget matrix $\mathbf{G}_n$ and thus the dimension of $\mathbf{R}_f$ depend on ℓ only logarithmically via the recursive packing technique, which yields $\|\sigma_f\| \le p(\lambda)\cdot\log\ell$ for a fixed polynomial p. □

A.4 Proof of Theorem 4

Proof. We reduce the unforgeability of FHS to the $\mathsf{SIS}_{q,n,m,\beta}$ problem. Given a random matrix $\mathbf{A}^* \in \mathbb{Z}_q^{n\times m}$ from a SIS challenger, we construct an algorithm $\mathcal{B}$ that simulates the security experiment for an adversary $\mathcal{A}$ and extracts a SIS solution from any successful forgery. The algorithm $\mathcal{B}$ first guesses which type of forgery (Type I or Type II) the adversary will produce.

Type I forgery (fresh tag τ^*). The algorithm $\mathcal{B}$ sets up the public key using the partitioning technique. It samples $\mathbf{U},\mathbf{U}_0,\mathbf{U}_1 \leftarrow\$ \{-1,1\}^{m\times m}$, $\mathbf{R},\mathbf{R}' \leftarrow\$ \{-1,1\}^{m\times m}$, and the partition hash parameters $(\mathbf{z},\boldsymbol{\alpha},\boldsymbol{\beta})$. It sets $\mathbf{A} = \mathbf{A}^*\mathbf{U}$, $\mathbf{D}_k = \mathbf{A}^*\mathbf{U}_k$ for $k \in \{0,1\}$, $\mathbf{B} = \mathbf{A}^*\mathbf{R} + \mathbf{E}$, and $\mathbf{B}' = \mathbf{A}^*\mathbf{R}' + \mathbf{E}'$, where $\mathbf{E},\mathbf{E}'$ encode the partition hash parameters. By Lemma 2.6 (leftover hash lemma), $(\mathbf{A},\mathbf{D}_0,\mathbf{D}_1,\mathbf{B},\mathbf{B}')$ are statistically close to uniform. The element $\mathbf{z} = \mathbf{A}^*\cdot\mathbf{r}_0$ for $\mathbf{r}_0 \leftarrow\$ \{0,1\}^m$ is computed honestly.

For each signing query (τ,μ_i,i) with $h_{\mathbf{z},\boldsymbol{\alpha},\boldsymbol{\beta}}(\tau) \neq \mathbf{0}$, the algorithm $\mathcal{B}$ runs $\mathbf{R}_\tau \leftarrow$ TrapEval to obtain a $\mathbf{G}_{\nu n}$-trapdoor for $\mathbf{A}_\tau = [\mathbf{A}^* \mid \mathbf{A} + \mathbf{B}_\tau]$. Using SampleDO, it

computes pre-signatures satisfying (1) and then applies Unpack to derive σ_i. The statistical closeness of $\mathsf{SampleDO}$ and $\mathsf{SampleLeft}$ (for the common Gaussian parameter s) makes sure that the signatures are indistinguishable from honestly generated ones. If $h_{\mathbf{z},\boldsymbol{\alpha},\boldsymbol{\beta}}(\tau)=\mathbf{0}$, the simulator aborts.

When $\mathcal{A}$ outputs a Type I forgery $(\tau^*,\sigma^*_{f^*},\mu^*,f^*)$ with $\tau^*\neq\tau_j$ for all queried τ_j, the event $h_{\mathbf{z},\boldsymbol{\alpha},\boldsymbol{\beta}}(\tau^*)=\mathbf{0}$ occurs with probability at least $0.5/q^u$, which is non-negligible. In this case, $\mathbf{B}_{\tau^*}=\mathbf{A}^*\mathbf{R}_{\tau^*}$ (the $\mathbf{E}_{\tau^*}\mathbf{G}_{\nu n}$ term vanishes) and the verification equation gives

$$\mathbf{A}^*\big(\mathbf{R}^*_{f^*,1}+(\mathbf{U}+\mathbf{R}_{\tau^*})\mathbf{R}^*_{f^*,2}-\mathbf{U}^*_{f^*}\big)=(k+\mu^*)\mathbf{G}_n,$$

where $\mathbf{D}^*_{f^*}=\mathbf{A}^*\mathbf{U}^*_{f^*}+k\mathbf{G}_n$ by the homomorphic structure (Lemma 4.2 of [27]). Since $\mathbf{R}_{\tau^*}$ is information-theoretically hidden from $\mathcal{A}$, the vector $\mathbf{w}=\mathbf{R}^*_{f^*,1}+(\mathbf{U}+\mathbf{R}_{\tau^*})\mathbf{R}^*_{f^*,2}-\mathbf{U}^*_{f^*}$ is non-zero with overwhelming probability. If $k+\mu^*=0$, then $\mathbf{A}^*\mathbf{w}=\mathbf{0}$ directly yields a SIS solution; otherwise, $\mathbf{w}'=\mathbf{w}\cdot\mathbf{T}_{\mathbf{G}_n}$ satisfies $\mathbf{A}^*\mathbf{w}'=\mathbf{0}$, with $\|\mathbf{w}'\|\leq\beta$.

Type II forgery (previously queried tag $\tau^*=\tau_j$, but $\mu^*\neq f^*(\boldsymbol{\mu}_j)$). The algorithm $\mathcal{B}$ randomly guesses the target tag τ^* and sets $\mathbf{A}=\mathbf{A}^*\mathbf{U}-\mathbf{B}_{\tau^*}$ so that $\mathbf{A}_{\tau^*}=[\mathbf{A}^*\,|\,\mathbf{A}^*\mathbf{U}]$. It samples pre-signatures $[\mathbf{R}^\top_{b,1}\,|\,\mathbf{R}^\top_{b,2}]^\top\leftarrow(\mathcal{D}_{\mathbb{Z}^m,s})^{2m}$ for $b\in\{0,1\}$ and defines $\mathbf{D}_b=[\mathbf{A}^*\,|\,\mathbf{A}^*\mathbf{U}]\cdot[\mathbf{R}^\top_{b,1}\,|\,\mathbf{R}^\top_{b,2}]^\top-[b\mathbf{I}_n\,|\cdots|\,b\mathbf{I}_n]\mathbf{G}_n$. We already know $(\mathbf{A},\mathbf{D}_0,\mathbf{D}_1)$ are statistically close to uniform. Signing queries for τ^* are answered using Unpack on the pre-generated pre-signatures. Other tags cause an abort.

For a Type II forgery, $\mathcal{B}$ also computes the honest evaluated signature $\mathbf{R}_{f^*}$ from the pre-signatures. Subtracting the forged verification equation from the honest one yields

$$\mathbf{A}^*\big(\mathbf{R}_{f^*,1}-\mathbf{R}^*_{f^*,1}+\mathbf{U}(\mathbf{R}_{f^*,2}-\mathbf{R}^*_{f^*,2})\big)=\big(f^*(\boldsymbol{\mu}_j)-\mu^*\big)\mathbf{G}_n.$$

Since $f^*(\boldsymbol{\mu}_j)\neq\mu^*$ and the pre-signatures are sampled from a continuous Gaussian distribution hidden from $\mathcal{A}$, the left-hand vector is non-zero with overwhelming probability. The resulting non-zero short vector (or its product with $\mathbf{T}_{\mathbf{G}_n}$) yields a valid SIS solution.

The guess of forgery type succeeds with probability $1/2$, and the partition succeeds with non-negligible probability. The total advantage of $\mathcal{B}$ in solving SIS is therefore a non-negligible fraction of $\mathcal{A}$'s advantage, which completes our reduction. □

A.5 Proof of Theorem 5

Proof. Since $\mathsf{Verify}(vk,\tau,\sigma_f,f)=1$, we have $\mathbf{A}_\tau\mathbf{R}_f=\mathbf{D}_f+y\mathbf{G}_n$ and $\|\mathbf{R}_f\|_\infty\leq B\leq\beta_{\max}$. The Hide procedure forms

$$\mathbf{H}_f=[\mathbf{A}_\tau\,|\,\mathbf{D}_f+(y-1)\mathbf{G}_n]\in\mathbb{Z}_q^{n\times 3m}.$$

Observe that $\mathbf{D}_f+(y-1)\mathbf{G}_n=\mathbf{A}_\tau\mathbf{R}_f+(-1)\mathbf{G}_n$, so the second block of $\mathbf{H}_f$ has the form $\mathbf{A}_\tau\mathbf{R}_f+h\mathbf{G}_n$ with $h=-1\neq 0$. This means $\mathbf{R}_f$ is a valid $\mathbf{G}_n$-trapdoor for $\mathbf{H}_f$ with non-zero tag $h=-1$ (in the sense of Definition 3.4 of [27]).

The $\mathsf{SampleRight}$ algorithm, on input $(\mathbf{H}_f,\mathbf{R}_f,\mathbf{z})$, exploits this trapdoor to sample a short vector $\tilde{\sigma}_f\in\mathbb{Z}^{3m}$ satisfying $\mathbf{H}_f\cdot\tilde{\sigma}_f=\mathbf{z}$ and $\|\tilde{\sigma}_f\|_\infty\leq\tilde{\beta}_{\max}$. Both conditions are exactly those checked by $\mathsf{HVerify}$, so $\mathsf{HVerify}(vk,\tau,\tilde{\sigma}_f,y,f)=1$. □

A.6 Proof of Theorem 6

Proof. Suppose an adversary $\mathcal{A}$ outputs a forgery $(f^*,y',\tilde{\sigma}')$ s.t. $\mathsf{HVerify}(vk,\tau^*,\tilde{\sigma}',y',f^*)=1$ but $y'\neq y=f^*(\boldsymbol{\mu})$. This means $\mathbf{H}_{f^*}\cdot\tilde{\sigma}'=\mathbf{z}$ where $\mathbf{H}_{f^*}=[\mathbf{A}_\tau\,|\,\mathbf{D}_{f^*}+(y'-1)\mathbf{G}_n]$. The reduction $\mathcal{B}$ also knows the honest evaluated signature $\mathbf{R}_f$ certifying $y=f^*(\boldsymbol{\mu})$. It computes $\tilde{\sigma}\leftarrow\mathsf{Hide}(vk,y,\sigma_f)$, obtaining $\mathbf{H}'_f\cdot\tilde{\sigma}=\mathbf{z}$ where $\mathbf{H}'_f=[\mathbf{A}_\tau\,|\,\mathbf{D}_{f^*}+(y-1)\mathbf{G}_n]$.

Write $\tilde{\sigma}'=(\mathbf{r}_1',\mathbf{r}_2')\in\mathbb{Z}^{2m}\times\mathbb{Z}^m$ and $\tilde{\sigma}=(\mathbf{r}_1,\mathbf{r}_2)\in\mathbb{Z}^{2m}\times\mathbb{Z}^m$. Then:

$$\mathbf{A}_\tau\mathbf{r}_1'+(\mathbf{D}_{f^*}+(y'-1)\mathbf{G}_n)\mathbf{r}_2'=\mathbf{z},$$
$$\mathbf{A}_\tau\mathbf{r}_1+(\mathbf{D}_{f^*}+(y-1)\mathbf{G}_n)\mathbf{r}_2=\mathbf{z}.$$

Subtracting and rearranging, we will have:

$$\mathbf{A}_\tau(\mathbf{r}_1'-\mathbf{r}_1)+\mathbf{D}_{f^*}(\mathbf{r}_2'-\mathbf{r}_2)+(y'-1)\mathbf{G}_n\mathbf{r}_2'-(y-1)\mathbf{G}_n\mathbf{r}_2=\mathbf{0}.$$

Since $y'\neq y$, the $\mathbf{G}_n$-components do not cancel. Expanding $\mathbf{A}_\tau=[\mathbf{A}^*\,|\,\mathbf{A}^*\mathbf{U}]$ (in the Type II simulation) and collecting terms involving $\mathbf{A}^*$, one extracts a short non-zero vector $\mathbf{w}$ with $\mathbf{A}^*\mathbf{w}=c\cdot\mathbf{G}_n$ for some $c\neq 0$. Multiplying by $\mathbf{T}_{\mathbf{G}_n}$ yields $\mathbf{A}^*(\mathbf{w}\cdot\mathbf{T}_{\mathbf{G}_n})=\mathbf{0}$. The norm bound $\|\mathbf{w}\cdot\mathbf{T}_{\mathbf{G}_n}\|_\infty\leq m\cdot\tilde{\beta}_{\max}^2\cdot\sqrt{5}\leq\beta_{\mathsf{SIS}}$ holds for appropriately chosen parameters, and the non-zeroness follows from the entropy argument on $\mathbf{r}_0$ (cf. the proof of Theorem 3.1 in [26]): $H_\infty(\mathbf{r}_0\,|\,\mathbf{z}=\mathbf{A}^*\mathbf{r}_0)\geq m-n\log q=\omega(\log\lambda)$, so the probability that the extracted vector is zero is negligible. □

A.7 Proof of Theorem 7

Proof. We need to show that $\mathsf{Hide}(vk,y,\sigma_f)\overset{\text{stat}}{\approx}\mathsf{Sim}(sk,vk,\tau,y,f)$.

The Hide algorithm. Given $\sigma_f=(\mathbf{R}_f,y,f)$ with $\mathbf{A}_\tau\mathbf{R}_f=\mathbf{D}_f+y\mathbf{G}_n$, the Hide procedure forms $\mathbf{H}_f=[\mathbf{A}_\tau\,|\,\mathbf{D}_f+(y-1)\mathbf{G}_n]$ where the second block equals $\mathbf{A}_\tau\mathbf{R}_f+(-1)\mathbf{G}_n$, so $\mathbf{R}_f$ is a $\mathbf{G}_n$-trapdoor for $\mathbf{H}_f$ with tag $h=-1\neq 0$. It outputs $\tilde{\sigma}_f\leftarrow\mathsf{SampleRight}(\mathbf{H}_f,\mathbf{R}_f,\mathbf{z})$.

The Sim algorithm. Given $(sk=\mathbf{T}_{\mathbf{A}^*},vk,\tau,y,f)$, the simulator recomputes $\mathbf{D}_f$ from the public parameters and the circuit f, and forms the same matrix $\mathbf{H}_f=[\mathbf{A}_\tau\,|\,\mathbf{D}_f+(y-1)\mathbf{G}_n]$. Writing $\mathbf{A}_\tau=[\mathbf{A}^*\,|\,\mathbf{A}+\mathbf{B}_\tau]$, we have

$$\mathbf{H}_f=[\mathbf{A}^*\,|\,\mathbf{A}+\mathbf{B}_\tau\,|\,\mathbf{D}_f+(y-1)\mathbf{G}_n].$$

Since $\mathbf{T}_{\mathbf{A}^*}$ is a short basis for $\Lambda_q^\perp(\mathbf{A}^*)$, the simulator invokes $\tilde{\sigma}_f\leftarrow\mathsf{SampleLeft}(\mathbf{A}^*,[\mathbf{A}+\mathbf{B}_\tau\,|\,\mathbf{D}_f+(y-1)\mathbf{G}_n],\mathbf{T}_{\mathbf{A}^*},\mathbf{z},\tilde{s})$, which uses the trapdoor of the *left* block $\mathbf{A}^*$ to sample from the coset $\Lambda_q^{\mathbf{z}}(\mathbf{H}_f)$.

The key observation is that both SampleRight and SampleLeft produce outputs drawn from distributions within negligible statistical distance of the discrete Gaussian $\mathcal{D}_{\Lambda_q^{\mathbf{z}}(\mathbf{H}_f),\tilde{s}}$ over the same coset $\Lambda_q^{\mathbf{z}}(\mathbf{H}_f)=\{\mathbf{v}\in\mathbb{Z}^{3m}:\mathbf{H}_f\cdot\mathbf{v}=\mathbf{z}\bmod q\}$, provided the Gaussian parameter $\tilde{s}$ is sufficiently large. Specifically:

- SampleRight requires $\tilde{s}\geq\|\mathbf{R}_f\|_\infty\cdot p_{\text{extra}}(n,m,\log q)$ and uses the $\mathbf{G}_n$-trapdoor induced by $\mathbf{R}_f$ with tag $h=-1\neq 0$.
- SampleLeft requires $\tilde{s}>\|\widetilde{\mathbf{T}}_{\mathbf{A}^*}\|\cdot\omega(\sqrt{\log 3m})$ and uses the basis $\mathbf{T}_{\mathbf{A}^*}$ of $\Lambda_q^\perp(\mathbf{A}^*)$.

By Lemma 6.4 of [26] (adapted from [91]), for any $(\mathbf{A}^*,\mathsf{td})\in\mathsf{TrapGen}(1^n,1^m,q)$, any $\mathbf{z}\in\mathbb{Z}_q^n$, any $\mathbf{R}_f$ with $\|\mathbf{R}_f\|_\infty\leq\beta_{\max}$, and any $h\neq 0$, we have:

$$\mathsf{SampleLeft}(\mathbf{H}_f,\mathsf{td},\mathbf{z})\overset{\text{stat}}{\approx}\mathsf{SampleRight}(\mathbf{H}_f,\mathbf{R}_f,\mathbf{z}),$$

where the statistical distance is $\mathsf{negl}(n)$. Setting $\tilde{s}=\tilde{\beta}_{\max}=\beta_{\max}\cdot p_{\text{extra}}(n,m,\log q)$ to satisfy both requirements simultaneously, we obtain

$$\mathsf{Hide}(vk,y,\sigma_f)\overset{\text{stat}}{\approx}\mathsf{Sim}(sk,vk,\tau,y,f).$$

Note that Sim depends only on (sk,vk,τ,y,f) and not on the underlying data $\boldsymbol{\mu}$ or the original signature σ_f. Therefore, the hidden signature $\tilde{\sigma}_f$ reveals no information about $\boldsymbol{\mu}$ beyond $y=f(\boldsymbol{\mu})$. Furthermore, the tag component in vk is $\mathbf{A}_\tau=[\mathbf{A}^*\,|\,\mathbf{A}+\mathbf{B}_\tau]$ which depends only on τ and the public parameters, not on $\{\mu_i\}$. □

A.8 Formal Proofs for Phoenix Security

Proof of Theorem 8. We proceed by constructions of Recover and Sim, and then bounding the advantage of $\mathcal{A}$ via a sequence of hybrid games G_0,G_1,G_2,G_3 such that $G_0 \equiv \mathsf{Exp}^{\text{du-mb-BPRIV},0}_{\mathcal{A},\text{Phoenix},\text{Sim},\text{Recover}}(\lambda)$ and $G_3 \equiv \mathsf{Exp}^{\text{du-mb-BPRIV},1}_{\mathcal{A},\text{Phoenix},\text{Sim},\text{Recover}}(\lambda)$.

On input (BB,BB_0,BB_1), define the selection function π as follows. For each entry $b=(id,upk,c,\sigma,s)$ on the adversarially produced board BB:

- If there exists a unique index j such that $(id,(upk,b_1^{(j)})) \in BB_1$ and $b=b_1^{(j)}$ (i.e., b is the $\beta=1$ version of the j-th honest query), set π to select the corresponding entry $b_0^{(j)}$ from BB_0.
- Otherwise (i.e., b was cast by a dishonest voter or injected by $\mathcal{A}$), set π to copy b verbatim.

Output $BB' \leftarrow \bar{\pi}(BB_0)$. By construction, BB' contains the $\beta=0$ ballot for every honest voter whose $\beta=1$ ballot appears in BB, and retains all adversarial entries unchanged.

On input (pd,pbb,r) where $pbb = \mathsf{Publish}(BB)$ and r is the tally result obtained from BB', the simulator constructs a simulated tally proof Π' by invoking the zero-knowledge simulator $\mathsf{Sim}_{\mathsf{ZKP}}$ for ZKP on the statement (pk,pbb,r), producing $\Pi' \leftarrow \mathsf{Sim}_{\mathsf{ZKP}}(pp,(pk,pbb,r))$.

Game G_0. This is the real du-mb-BPRIV experiment with $\beta = 0$. The oracle $\mathcal{O}_{\mathsf{voteLR}}(id,v_0,v_1)$ computes both $b_0 \leftarrow \mathsf{Vote}(id,v_0,pk,upk,usk)$ and $b_1 \leftarrow \mathsf{Vote}(id,v_1,pk,upk,usk)$, appends $(id,(upk,b_i))$ to BB_i for $i \in \{0,1\}$, records voter state from $\beta = 0$, and returns b_0 to $\mathcal{A}$ (via $\mathcal{O}_{\mathsf{board}}$, which publishes BB_0). The oracle $\mathcal{O}_{\mathsf{tally}}$ computes $(r,\Pi) \leftarrow \mathsf{Tally}(BB,pd,sd)$ on the adversarially-chosen board BB and returns (r,Π); recall that Tally first retains, for each credential, only its most recent entry in BB (yielding BB^*) before verification and aggregation. The oracle $\mathcal{O}_{\mathsf{verify}}(id)$ runs $\mathsf{VerifyVote}(b_0,pbb)$ using the $\beta=0$ ballot and the published board.

Game G_1. G_1 is identical to G_0 except that in $\mathcal{O}_{\mathsf{voteLR}}$, the ciphertext component of the ballot *shown to the adversary* (i.e., on BB_β for $\beta=0$ in G_0) is replaced: for each honest query $j \in [q_v]$, we now encrypt $v_1^{(j)}$ instead of $v_0^{(j)}$, i.e., $c^{(j)} \leftarrow \mathsf{BFV.Enc}(pk,v_1^{(j)})$. The ZKP $\sigma^{(j)}$ and FHS signature $s^{(j)}$ are still computed consistently with respect to this new ciphertext and its actual witness $(v_1^{(j)},\mathbf{s}_1,\mathbf{e},\mathbf{s}_2)$. The tally oracle $\mathcal{O}_{\mathsf{tally}}$ and verify oracle $\mathcal{O}_{\mathsf{verify}}$ remain as in G_0: tallying uses BB' (recovered via Recover, which maps honest entries back to BB_0, i.e., encryptions of $v_0^{(j)}$) and verification uses the $\beta=0$ ballot states.

We bound $|\Pr[G_0=1]-\Pr[G_1=1]|$ by a standard hybrid argument over the q_v voting queries. Define sub-hybrids $G_{0,0},G_{0,1},\ldots,G_{0,q_v}$ where $G_{0,i}$ encrypts v_1 for the first i honest queries and v_0 for the remaining q_v-i queries; thus $G_{0,0} \equiv G_0$ and $G_{0,q_v} \equiv G_1$.

For the transition from $G_{0,i-1}$ to $G_{0,i}$, we construct a reduction $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ against IND-CPA of BFV. The reduction $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ receives pk from the IND-CPA challenger, uses it as the election public key, and simulates the experiment for $\mathcal{A}$. For all honest queries $j \neq i$, it encrypts according to the hybrid specification (i.e., $v_1^{(j)}$ for $j<i$ and $v_0^{(j)}$ for $j>i$). For the i-th query $(id_i,v_0^{(i)},v_1^{(i)})$, $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ submits $(v_0^{(i)},v_1^{(i)})$ to the IND-CPA challenger and receives a challenge ciphertext c^*. It then computes $\sigma^{(i)} \leftarrow \mathsf{ZKP.Prove}(pp,(c^*,upk_i),w_b)$ using the witness corresponding to whichever plaintext was encrypted (which $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ does not know, but can compute for *both* $v_0^{(i)},v_1^{(i)}$ by running BFV.Enc itself for the ZKP

witness and using c^* only as the statement—more precisely, $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ uses the randomness it chose to produce the ZKP for whichever value the challenger encrypted).

$\mathcal{B}^{(i)}_{\mathsf{cpa}}$ does not know the bit b chosen by the IND-CPA challenger and thus cannot produce the ZKP witness directly for c^*. We address this by a further sub-hybrid: $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ first produces $c_0 \leftarrow \mathsf{BFV.Enc}(pk, v_0^{(i)}; r_0)$ and $c_1 \leftarrow \mathsf{BFV.Enc}(pk, v_1^{(i)}; r_1)$ itself (retaining both witnesses), submits $(v_0^{(i)}, v_1^{(i)})$ to the IND-CPA oracle to receive c^*, and uses the IND-CPA security *a second time* to argue that replacing c_b with c^* (which encrypts the same plaintext under the same key) is indistinguishable. Since $\mathsf{BFV.Enc}$ is IND-CPA secure, c^* and c_b are identically distributed given pk and the plaintext, so $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ can use witness r_b for the ZKP on c_b, then appeal to a re-randomization argument (or simply embed c^* directly since the ZKP proves knowledge of *some* valid witness, and completeness ensures one exists). In fact, $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ computes both proofs $\sigma_0^{(i)}, \sigma_1^{(i)}$ for c_0, c_1 respectively, and uses the one matching the IND-CPA challenge in $G_{0,i-1}$ versus $G_{0,i}$. The FHS signature $s^{(i)}$ is then computed on the pair $(c_b, \sigma_b^{(i)})$ via $\mathsf{FHS.Sign}(usk_i, vk_i, \tau_i, (c_b, \sigma_b^{(i)}), 1)$ using usk_i and the per-voter (vk_i, τ_i) parsed from upk_i, all of which $\mathcal{B}^{(i)}_{\mathsf{cpa}}$ possesses.

Since $\mathcal{A}$'s view differs between $G_{0,i-1}$ and $G_{0,i}$ only in whether the i-th ciphertext encrypts $v_0^{(i)}$ or $v_1^{(i)}$, we obtain:

$$|\Pr[G_{0,i-1}=1] - \Pr[G_{0,i}=1]| \leq \mathsf{Adv}^{\mathsf{ind\text{-}cpa}}_{\mathcal{B}^{(i)}_{\mathsf{cpa}},\mathsf{BFV}}(\lambda).$$

Summing over $i \in [q_v]$ by the triangle inequality:

$$|\Pr[G_0=1] - \Pr[G_1=1]| \leq q_v \cdot \mathsf{Adv}^{\mathsf{ind\text{-}cpa}}_{\mathcal{B}_{\mathsf{cpa}},\mathsf{BFV}}(\lambda),$$

where $\mathsf{Adv}^{\mathsf{ind\text{-}cpa}}_{\mathcal{B}_{\mathsf{cpa}},\mathsf{BFV}} = \max_i \mathsf{Adv}^{\mathsf{ind\text{-}cpa}}_{\mathcal{B}^{(i)}_{\mathsf{cpa}},\mathsf{BFV}}$.

Game G_2. G_2 is identical to G_1 except that for each honest voting query $j \in [q_v]$, the zero-knowledge proof $\sigma^{(j)}$ accompanying the ciphertext $c^{(j)} = \mathsf{BFV.Enc}(pk, v_1^{(j)})$ on the board visible to $\mathcal{A}$ is replaced by a simulated proof: $\sigma^{(j)} \leftarrow \mathsf{Sim}_{\mathsf{ZKP}}(pp, (c^{(j)}, upk_j))$, which is produced without knowledge of the witness $(v_1^{(j)}, \mathbf{s}_1, \mathbf{e}, \mathbf{s}_2)$.

Define $G_{1,0}, \ldots, G_{1,q_v}$ where $G_{1,i}$ simulates the ZKP for the first i queries and produces real proofs for queries $i+1, \ldots, q_v$. For the i-th transition, we construct $\mathcal{B}^{(i)}_{\mathsf{zk}}$ that plays the ZK distinguishing game: it receives either a real proof or a simulated proof for the i-th statement and embeds it into $\mathcal{A}$'s view. The FHS signature $s^{(j)} \leftarrow \mathsf{FHS.Sign}(usk_j, vk_j, \tau_j, (c^{(j)}, \sigma^{(j)}), 1)$ is recomputed consistently—since $\mathsf{FHS.Sign}$ treats $(c^{(j)}, \sigma^{(j)})$ as an opaque message and the reduction holds usk_j together with the per-voter (vk_j, τ_j), the signature can be produced on whichever $(c^{(j)}, \sigma^{(j)})$ pair is determined by the ZK game.

$\mathsf{FHS.Sign}$ is a deterministic-given-randomness PPT algorithm, and its output distribution depends on its input (c, σ) only through the value of that input. Since $(c^{(j)}, \sigma^{(j)}_{\mathsf{real}})$ and $(c^{(j)}, \sigma^{(j)}_{\mathsf{sim}})$ are computationally indistinguishable, the resulting signatures $s^{(j)}$ are equally indistinguishable. Thus $|\Pr[G_1=1] - \Pr[G_2=1]| \leq q_v \cdot \mathsf{Adv}^{\mathsf{zk}}_{\mathcal{B}_{\mathsf{zk}},\mathsf{ZKP}}(\lambda)$.

Game G_3. G_3 is identical to G_2 except that $\mathcal{O}_{\mathsf{tally}}$ no longer tallies the adversarially-chosen board BB directly. Instead, it computes $BB' \leftarrow \mathsf{Recover}(BB, BB_0, BB_1)$ and then $(r, \Pi) \leftarrow \mathsf{Tally}(BB', pd, sd)$, returning $(r, \mathsf{Sim}(pd, \mathsf{Publish}(BB), r))$ where the tally proof is simulated.

We argue that G_2 and G_3 are indistinguishable unless $\mathcal{A}$ causes $\mathsf{Recover}$ to fail, i.e., $\mathcal{A}$ places on BB a ballot $b^*=(id^*,upk^*,c^*,\sigma^*,s^*)$ that (a) passes Valid, (b) upk^* belongs to an honest voter id^*, yet (c) $(c^*,\sigma^*)\neq(c_1^{(j)},\sigma_1^{(j)})$ for any honest ballot $b_1^{(j)}$ associated with id^* (across all queries j made for id^*, since Tally retains only the latest entry per credential in BB^*). In this case, parsing $upk^*=(vk^*,\tau^*)$, the signature s^* is a valid FHS signature under (vk^*,τ^*) on a message (c^*,σ^*) that was never signed by the legitimate key holder, constituting a forgery.

We construct a reduction $\mathcal{B}_{\mathsf{fhs}}$ against $\mathsf{EU\text{-}F\text{-}ACMA}$ of FHS. $\mathcal{B}_{\mathsf{fhs}}$ receives pk_{fhs} from the FHS challenger and embeds it as the credential setup for a randomly chosen honest voter id^* (guessing the target with probability $\geq 1/|\mathcal{H}|$, absorbed into the advantage term). $\mathcal{B}_{\mathsf{fhs}}$ samples a fresh tag $\tau^*\leftarrow\$\{0,1\}^{|\tau|}$, queries its KeyGen-derived vk^*, and sets $upk^*=(vk^*,\tau^*)$. For signing queries on behalf of id^*, $\mathcal{B}_{\mathsf{fhs}}$ forwards them to its FHS signing oracle under tag τ^*. If $\mathcal{A}$ produces a ballot with a valid FHS signature under (vk^*,τ^*) on a never-queried message (c^*,σ^*), $\mathcal{B}_{\mathsf{fhs}}$ extracts a Type II forgery (same tag τ^*, different signed message). Therefore:

$$|\Pr[G_2=1]-\Pr[G_3=1]|\leq|\mathcal{H}|\cdot\mathsf{Adv}^{\mathsf{euf\text{-}acma}}_{\mathcal{B}_{\mathsf{fhs}},\mathsf{FHS}}(\lambda).$$

Additionally, the switch from the real tally proof Π to the simulated proof as the $\mathsf{Sim}(pd,\mathsf{Publish}(BB),r)$ is justified by the zero-knowledge property of ZKP applied to the tally proof. Since $\mathcal{O}_{\mathsf{tally}}$ is invoked at most once, this contributes an additive $\mathsf{Adv}^{\mathsf{zk}}_{\mathsf{ZKP}}(\lambda)$ term (which we absorb into the existing $q_v\cdot\mathsf{Adv}^{\mathsf{zk}}$ term from the previous hop for cleanliness, noting $q_v\geq 1$).

Game G_3 to Exp^1. We now argue that $G_3\equiv\mathsf{Exp}^1$ up to the behavior of $\mathcal{O}_{\mathsf{verify}}$. In Exp^1, the oracle $\mathcal{O}_{\mathsf{verify}}(id)$ runs $\mathsf{VerifyVote}$ using st^1_{pre} (the $\beta=1$ ballot's hash fingerprint) against $pbb=\mathsf{Publish}(BB_1)$, which succeeds if the ballot was included. In G_3, the voter's pre-state is already from $\beta=1$ (since the board shown to $\mathcal{A}$ uses v_1-ciphertexts with simulated proofs, matching Exp^1), and pbb is derived from BB (controlled by $\mathcal{A}$).

Suppose $\mathcal{O}_{\mathsf{verify}}$ returns a different result in G_3 versus Exp^1. Since both games use the same board BB and the same ballot $b_1^{(j)}$ for the pre-state, this requires $H(\tilde{b}_1^{(j)})=H(\tilde{b}')$ for some $\tilde{b}'\neq\tilde{b}_1^{(j)}$ on the published board (where $\tilde{b}=\mathsf{remID}(b)$), or vice versa. Any such event yields a collision for H, so $|\Pr[G_3=1]-\Pr[\mathsf{Exp}^1=1]|\leq\mathsf{Adv}^{\mathsf{cr}}_{\mathcal{B}_{\mathsf{cr}},H}(\lambda)$.

Applying the triangle inequality across all four transitions:

$$\begin{aligned}\mathsf{Adv}^{\mathsf{du\text{-}mb\text{-}bpriv}}_{\mathcal{A},\mathsf{Phoenix},\mathsf{Sim}}(\lambda)&=|\Pr[\mathsf{Exp}^0=1]-\Pr[\mathsf{Exp}^1=1]|\\&\leq|\Pr[G_0=1]-\Pr[G_1=1]|+|\Pr[G_1=1]-\Pr[G_2=1]|\\&\quad+|\Pr[G_2=1]-\Pr[G_3=1]|+|\Pr[G_3=1]-\Pr[\mathsf{Exp}^1=1]|\\&\leq q_v\cdot\mathsf{Adv}^{\mathsf{ind\text{-}cpa}}_{\mathcal{B}_{\mathsf{cpa}},\mathsf{BFV}}(\lambda)+q_v\cdot\mathsf{Adv}^{\mathsf{zk}}_{\mathcal{B}_{\mathsf{zk}},\mathsf{ZKP}}(\lambda)\\&\quad+|\mathcal{H}|\cdot\mathsf{Adv}^{\mathsf{euf\text{-}acma}}_{\mathcal{B}_{\mathsf{fhs}},\mathsf{FHS}}(\lambda)+\mathsf{Adv}^{\mathsf{cr}}_{\mathcal{B}_{\mathsf{cr}},H}(\lambda)+\mathsf{negl}(\lambda),\end{aligned}$$

where the $\mathsf{negl}(\lambda)$ term accounts for the negligible probability that BFV decryption fails on correctly formed ciphertexts during recovery. Since $|\mathcal{H}|\leq q_v$ and we can absorb the factor into the advantage term by a standard guessing argument (the reduction guesses the target voter uniformly at random), the bound simplifies to the statement of the theorem as $\mathsf{Phoenix}$ satisfies $\mathsf{du\text{-}mb\text{-}BPRIV}$. □

Proof of Theorem 9. Let $\mathcal{B}$ be a PPT adversary in $\mathsf{Exp}^{\mathsf{consis}}_{\mathcal{B},\mathsf{Phoenix},\mathcal{I}}(\lambda)$. $\mathcal{B}$ outputs BB such that $\forall b\in BB$, $\mathsf{ValidInd}(b,pk)=1$, i.e., $\mathsf{ZKP.Verify}(pp,(c,upk),\sigma)=1$ and $\mathsf{FHS.Verify}(vk,\tau,s,(c,\sigma))=1$ for each $b=(id,upk,c,\sigma,s)$ with $upk=(vk,\tau)$.

By knowledge soundness of ZKP under MSIS, for each accepting proof σ there exists an extractor $\mathcal{E}$ that, with probability $\geq 1-\mathsf{Adv}^{\mathsf{ks}}_{\mathsf{ZKP}}(\lambda)$, outputs a witness $(v_i,\mathbf{s}_1,\mathbf{e},\mathbf{s}_2)$ satisfying $c_i=\mathsf{BFV.Enc}(pk,v_i;\mathbf{s}_1,\mathbf{e},\mathbf{s}_2)$, $v_i\in\{0,1\}$, and $\|\mathbf{s}_1\|,\|\mathbf{e}\|,\|\mathbf{s}_2\|\leq\beta_{\mathsf{enc}}$.

By correctness of BFV, since each c_i encrypts v_i with noise bounded by β_{enc}, $\mathsf{BFV.Dec}(sk,c_i)=v_i$ with overwhelming probability. Hence $\mathsf{Extract}(b_i,sk)=(upk_i,v_i)$ for all $i\in[|BB|]$.

$\mathsf{Tally}(BB,sk)$ first retains only the latest entry per credential to form BB^*, then verifies each ballot, collects all ciphertexts $\{c_i\}_{i=1}^n$ from valid ballots into vbb, computes $c^*=\mathsf{BFV.Add}(vbb)$, and returns $r=\mathsf{BFV.Dec}(sk,c^*)$. By additive homomorphism: $\mathsf{BFV.Dec}(sk,c^*)=\sum_{i=1}^n v_i \pmod t$. On the other hand, $r'=\mathsf{Count}\circ\mathsf{Policy}(dbb)=\sum_{i=1}^n v_i$, where Policy applies the same latest-ballot-per-credential filter used by Tally. Since $n\leq|\mathcal{I}|$ and $v_i\in\{0,1\}$, we have $\sum v_i<t$ for any practical election size, so no modular reduction occurs and $r=r'$.

Therefore $\Pr[r\neq r'\wedge e]\leq\mathsf{Adv}^{\mathsf{ks}}_{\mathsf{ZKP}}(\lambda)+\mathsf{negl}(\lambda)$, where the negligible term bounds the probability of BFV decryption failure. □

Proof of Theorem 10. Let $\mathcal{B}$ be a PPT adversary in $\mathsf{Exp}^{\mathsf{corr}}_{\mathcal{B},\mathsf{Phoenix},\mathcal{I}}(\lambda)$. $\mathcal{B}$ outputs (id,v,BB) with $id\in\mathcal{I}$. The experiment retrieves (usk,upk) for id (with $upk=(vk,\tau)$ bundling the per-voter verification key and tag), computes $b\leftarrow\mathsf{Vote}(id,v,pk,upk,usk)$, and tests whether $e_1\wedge\neg\mathsf{Valid}(BB,b,pk)$. Parse $b=(id,upk,c,\sigma,s)$ where:

1. $c\leftarrow\mathsf{BFV.Enc}(pk,v)$ with honestly sampled randomness;
2. $\sigma\leftarrow\mathsf{ZKP.Prove}(pp,(c,upk),(v,\mathbf{s}_1,\mathbf{e},\mathbf{s}_2))$ with valid witness;
3. $s\leftarrow\mathsf{FHS.Sign}(usk,vk,\tau,(c,\sigma),1)$ with the registered trapdoor, verification key, and tag.

$\mathsf{Valid}(BB,b,pk)$ checks $e_1\wedge e_2\wedge e_3$:

e_1: The credential-consistency check. Conditioned on $e_1=1$ (which the experiment assumes), no conflicting (id,upk) pair exists in BB, so $e_1=1$ for the honest ballot.
e_2: By completeness of ZKP, $\mathsf{ZKP.Verify}(pp,(c,upk),\sigma)=1$ with probability $\geq 1-\mathsf{negl}(\lambda)$.
e_3: By signing correctness of FHS (Definition 4), $\mathsf{FHS.Verify}(vk,\tau,s,(c,\sigma))=1$ with probability $\geq 1-\mathsf{negl}(\lambda)$.

By a union bound, $\Pr[e_2=0\vee e_3=0]\leq\mathsf{negl}(\lambda)$. Hence $\Pr[e_1\wedge\neg(e_1\wedge e_2\wedge e_3)]=\Pr[e_1\wedge(\neg e_2\vee\neg e_3)]\leq\mathsf{negl}(\lambda)$. □

Proof of Theorem 11. We apply the composition theorem of Cortier et al. [9], which states: if a voting scheme $\mathcal{V}$ satisfies mb-BPRIV, strong correctness, and strong consistency, then for every PPT environment $\mathfrak{E}$ there exists a PPT simulator $\mathcal{S}$ such that $|\Pr[\mathsf{realexec}(\mathfrak{E}\|\mathcal{A}\|\mathcal{V})=1]-\Pr[\mathsf{idealexec}(\mathfrak{E}\|\mathcal{S}\|\mathcal{F}_{\mathrm{voting}}(\rho))=1]|\leq\mathsf{negl}(\lambda)$. It remains to verify the three preconditions for Phoenix:

1. du-mb-BPRIV implies mb-BPRIV, since du-mb-BPRIV strengthens the adversary by delaying verification until after tallying; any mb-BPRIV adversary is a fortiori a du-mb-BPRIV adversary with $\mathcal{H}_{\mathsf{check}}=\emptyset$. By Theorem 8, Phoenix satisfies du-mb-BPRIV.
2. Strong correctness by Theorem 10.
3. Strong consistency by Theorem 9.

The simulator $\mathcal{S}$ is constructed as follows: $\mathcal{S}$ plays the role of the adversary in the ideal world, receives $\mathsf{ack}(id)$ for each vote submitted by $\mathfrak{E}$, and upon receiving $r \leftarrow \rho(L)$ from $\mathcal{F}_{\text{voting}}(\rho)$, simulates the adversary's view (bulletin board and tally proof) using the Sim from the $\mathsf{du\text{-}mb\text{-}BPRIV}$ experiment. Strong correctness ensures $\mathcal{S}$ can simulate ballot acceptance; strong consistency ensures the simulated result equals $\rho(L)$; and $\mathsf{mb\text{-}BPRIV}$ ensures the simulated view is indistinguishable from the real one. The theorem follows. □

Proof of Theorem 12. We prove both experiments separately.

$\mathsf{Exp}^{\mathsf{ver\text{-}b}}$ (honest registrar, malicious board). Suppose $\mathcal{A}$ outputs (BB, r, Π) with $\mathsf{Verify}(BB,r,\Pi)=1$. We show that, except with negligible probability, there exists a valid decomposition $r=\rho(\{v_i^E\})\oplus_R\rho(\{v_i^A\})\oplus_R\rho(\{v_i^B\})$ with $|\{v_i^B\}|\leq|CU|$.

(Individual verifiability.) For each $id\in\mathcal{H}_{\mathsf{check}}$, $\mathsf{VerifyVote}$ checks $H(\tilde{b})\in\{h\,|\,(\cdot,h)\in pbb\}$, where $\tilde{b}=\mathsf{remID}(b)$. If $\mathcal{A}$ replaces ballot b with $b'\neq b$ while maintaining $H(\tilde{b})=H(\tilde{b}')$, this constitutes a collision in H, occurring with probability $\leq\mathsf{Adv}_H^{\mathsf{cr}}(\lambda)$. If $\mathcal{A}$ removes b entirely, the check fails and the voter detects the omission. Hence each checked voter's ballot is correctly included with probability $\geq 1-\mathsf{Adv}_H^{\mathsf{cr}}(\lambda)$.

(Universal verifiability.) Since $\mathsf{Verify}((pk, pbb, r), \pi) = 1$ and ZKP is knowledge-sound under MSIS, by the ABDLOP extractor [74] there exists sk' such that $r = \mathsf{BFV.Dec}(sk',\mathsf{BFV.Add}(\{c_i\}))$ where $\{c_i\}$ are the ciphertexts on pbb, except with probability $\leq\mathsf{Adv}_{\mathsf{ZKP}}^{\mathsf{ks}}(\lambda)$. Hence no adversary can claim an incorrect r for the published ciphertexts.

(Eligibility verifiability.) Every ballot $b\in BB$ accepted by Valid satisfies the relation $\mathsf{FHS.Verify}(vk,\tau,s,(c,\sigma))=1$, where $upk=(vk,\tau)$ is parsed from b. Since the registrar is honest, each (vk,τ) corresponds to a legitimately registered voter with a fresh per-voter tag $\tau\leftarrow\$\{0,1\}^{|\tau|}$. If $\mathcal{A}$ produces a valid signature under some $upk^*=(vk^*,\tau^*)$ without possessing the corresponding usk^*, this constitutes a forgery against FHS, occurring with probability $\leq\mathsf{Adv}_{\mathsf{FHS}}^{\mathsf{euf}}(\lambda)$. The number of adversarial ballots with valid signatures under corrupted credentials is at most $|CU|$, so $|\{v_i^B\}|\leq|CU|$.

Combining by a union bound: $\Pr[\mathsf{Exp}_{\mathcal{A},\mathsf{Phoenix}}^{\mathsf{ver\text{-}b}}(\lambda)=1]\leq\mathsf{Adv}_{\mathsf{FHS}}^{\mathsf{euf}}(\lambda)+\mathsf{Adv}_{\mathsf{ZKP}}^{\mathsf{ks}}(\lambda)+\mathsf{Adv}_H^{\mathsf{cr}}(\lambda)+\mathsf{negl}(\lambda)$.

$\mathsf{Exp}^{\mathsf{ver\text{-}r}}$ (honest board, malicious registrar). Here $\mathcal{A}$ controls the registrar and voters but the bulletin board is honest: ballots enter only via $\mathcal{O}_{\mathsf{cast}}(id,b)$, which runs $\mathsf{Valid}(BB,b,pk)$ before appending. Valid enforces: (i) credential-consistency, binding each id to a unique $upk=(vk,\tau)$; (ii) $\mathsf{ZKP.Verify}(pp,(c,upk),\sigma)=1$, making sure well-formedness; (iii) $\mathsf{FHS.Verify}(vk,\tau,s,(c,\sigma))=1$, making sure authentication. Since the board is honest, $\mathcal{A}$ cannot inject ballots bypassing validation. Knowledge soundness of ZKP ensures each accepted ciphertext encrypts a valid $v\in\{0,1\}$. The honest tally $\mathsf{Tally}(BB,sk)$ then correctly retains the latest ballot per credential and aggregates, so $r=\sum v_i$. Any decomposition $r=\rho(\{v_i^E\})\oplus_R\rho(\{v_i^A\})\oplus_R\rho(\{v_i^B\})$ with $|\{v_i^B\}|\leq|CU|$ is satisfiable by assigning corrupted voters' actual votes to $\{v_i^B\}$. Hence $\Pr[\mathsf{Exp}_{\mathcal{A},\mathsf{Phoenix}}^{\mathsf{ver\text{-}r}}(\lambda)=1]\leq\mathsf{negl}(\lambda)$. □

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